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(54) **STEAM CLEANING APPARATUS**

(71) Applicant: **SHARKNINJA OPERATING LLC**,
Needham, MA (US)

(72) Inventors: **Fariha Ahmed**, Needham, MA (US);
Joshua Redman, Needham, MA (US);
Peter Hutchinson, Needham, MA (US);
Dmitry Layevsky, Needham, MA (US);
Andrew Keene, Needham, MA (US);
Kai Xu, Suzhou (CN); **Ognjen**
Vrdoljak, Quebec (CA); **Queen Tan**,
Needham, MA (US); **Devin Feng**,
Suzhou (CN); **Junping Li**, Needham,
MA (US); **Ming Yao**, Suzhou (CN);
Tao Wang, Needham, MA (US); **Hui**
Zou, Needham, MA (US); **Yonggang**
Li, Needham, MA (US); **David T.**
Barker, Snaith (GB); **Helian Chen**,
Suzhou (CN); **Jibing Gu**, Needham,
MA (US); **Mingliang Qin**, Needham,
MA (US); **Zhengzhou Lu**, Needham,
MA (US); **Lei Liu**, Needham, MA (US)

(73) Assignee: **SHARKNINJA OPERATING LLC**,
Needham, MA (US)

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11/4069 (2013.01)

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Primary Examiner — David S Posigian

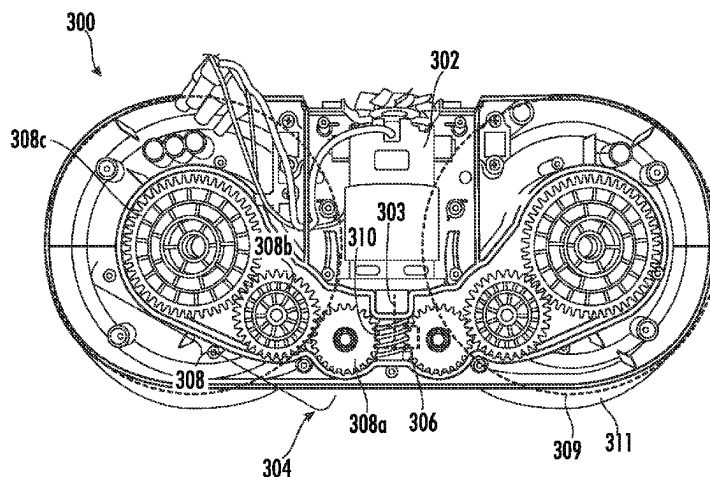
Assistant Examiner — Steven Huang

(74) *Attorney, Agent, or Firm* — Mintz, Levin, Cohn,
Ferris, Glovsky and Popeo, P.C.

(57) **ABSTRACT**

A steam cleaning apparatus may include a wand, a cleaning
assembly coupled to the wand, the cleaning assembly
including a steam generator, and a cleaning head pivotally
coupled to the wand. The cleaning head may include a
plurality of cleaning pads, a pad motor having a motor
driveshaft, a drivetrain configured to transfer rotational

(Continued)



motion of the motor driveshaft to each of the plurality of cleaning pads such that a rotation of the motor driveshaft causes a corresponding rotation in each of the plurality of cleaning pads, and a temperature sensor positioned proximate to the drivetrain, the temperature sensor being configured to measure a temperature of the drivetrain.

14 Claims, 19 Drawing Sheets

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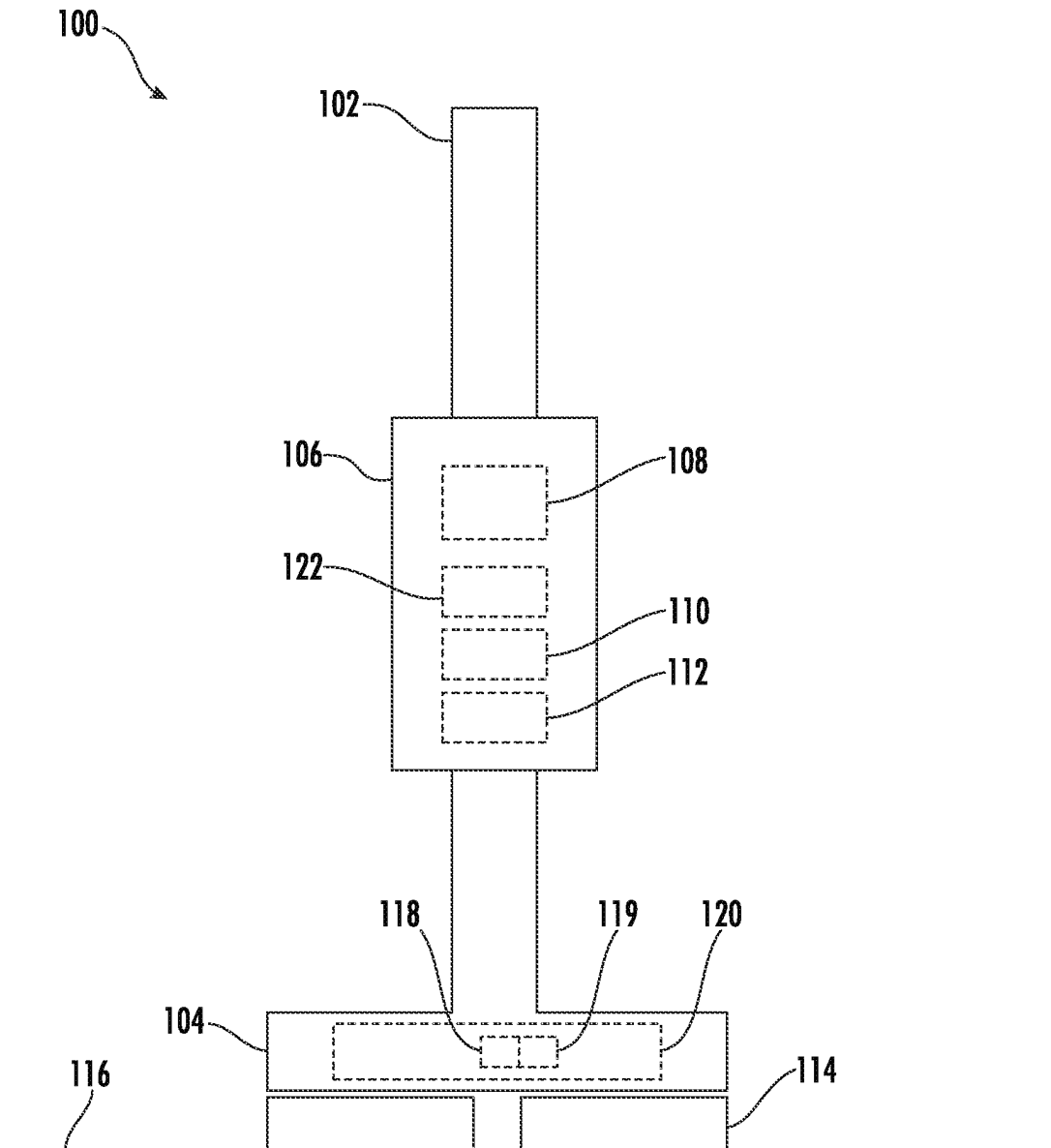
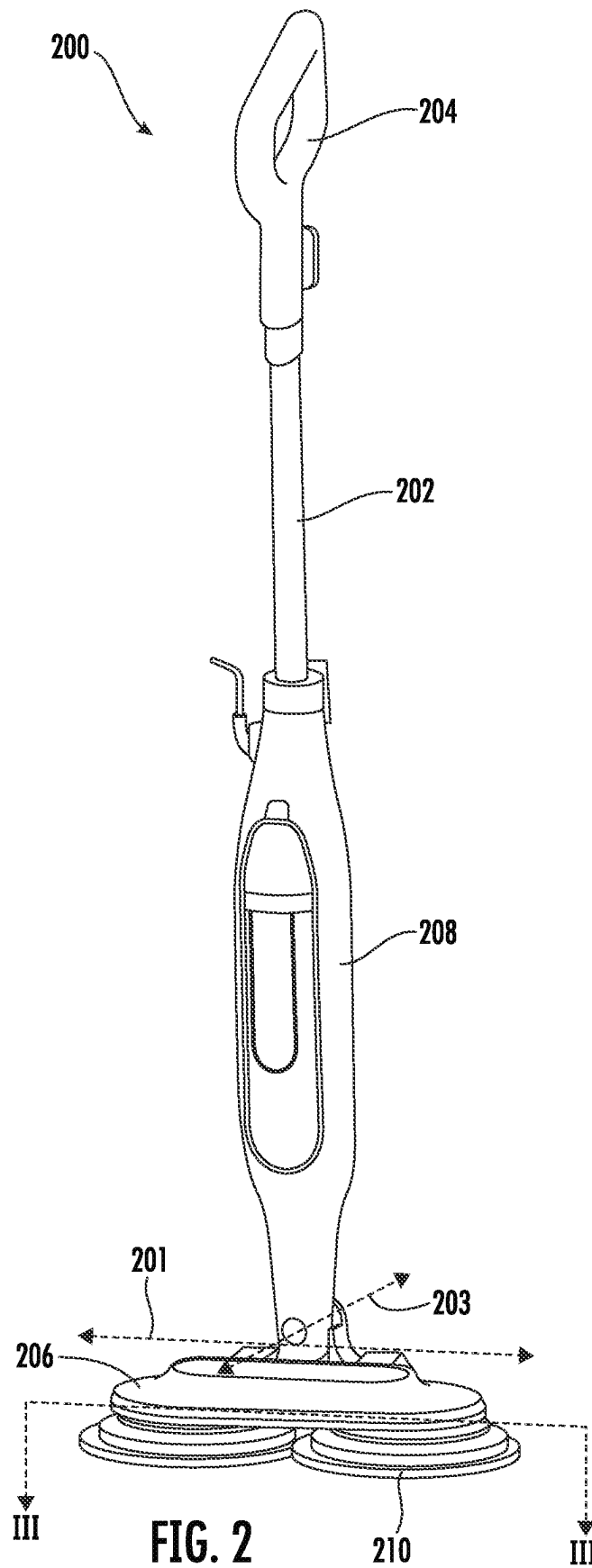


FIG. 1



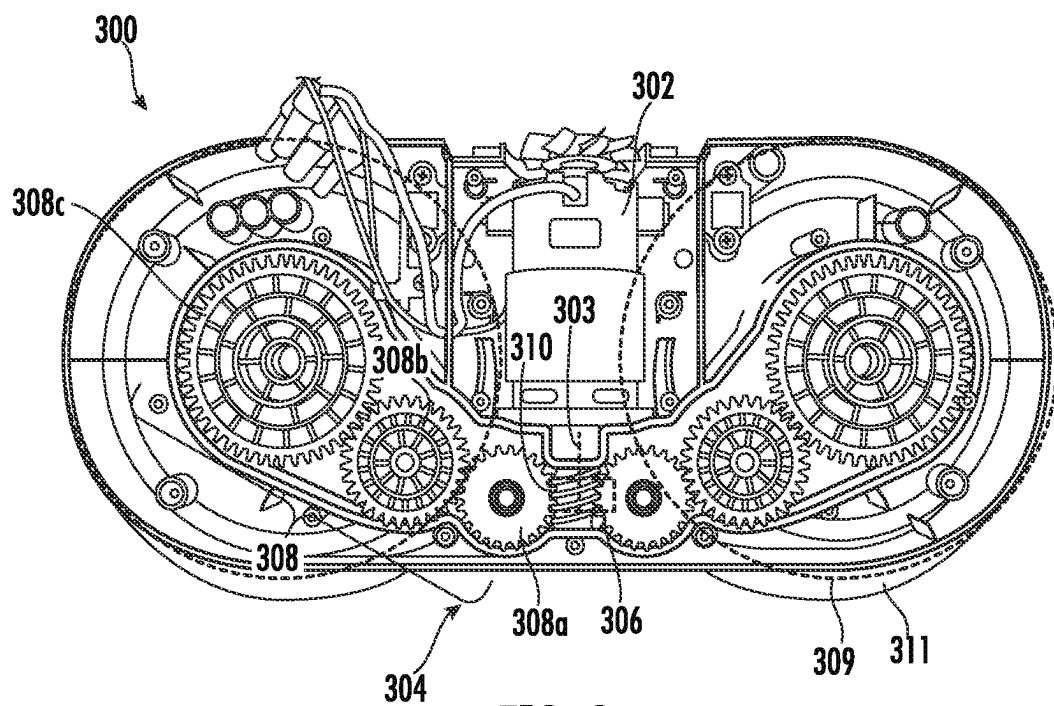


FIG. 3

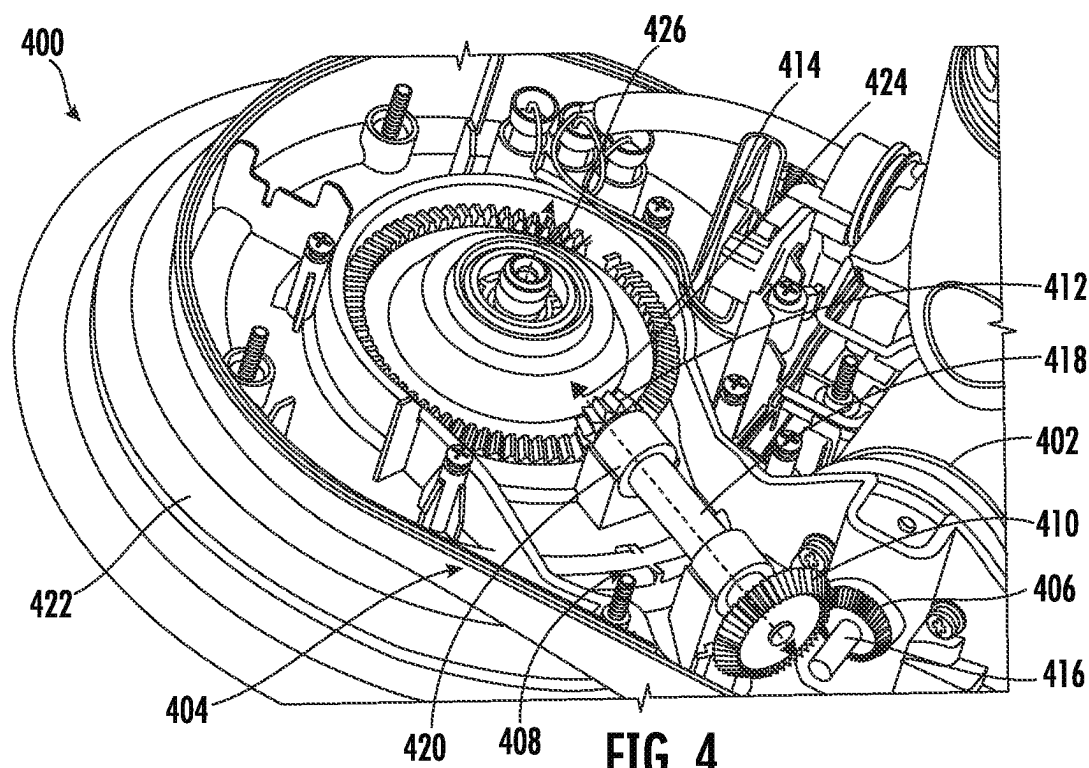
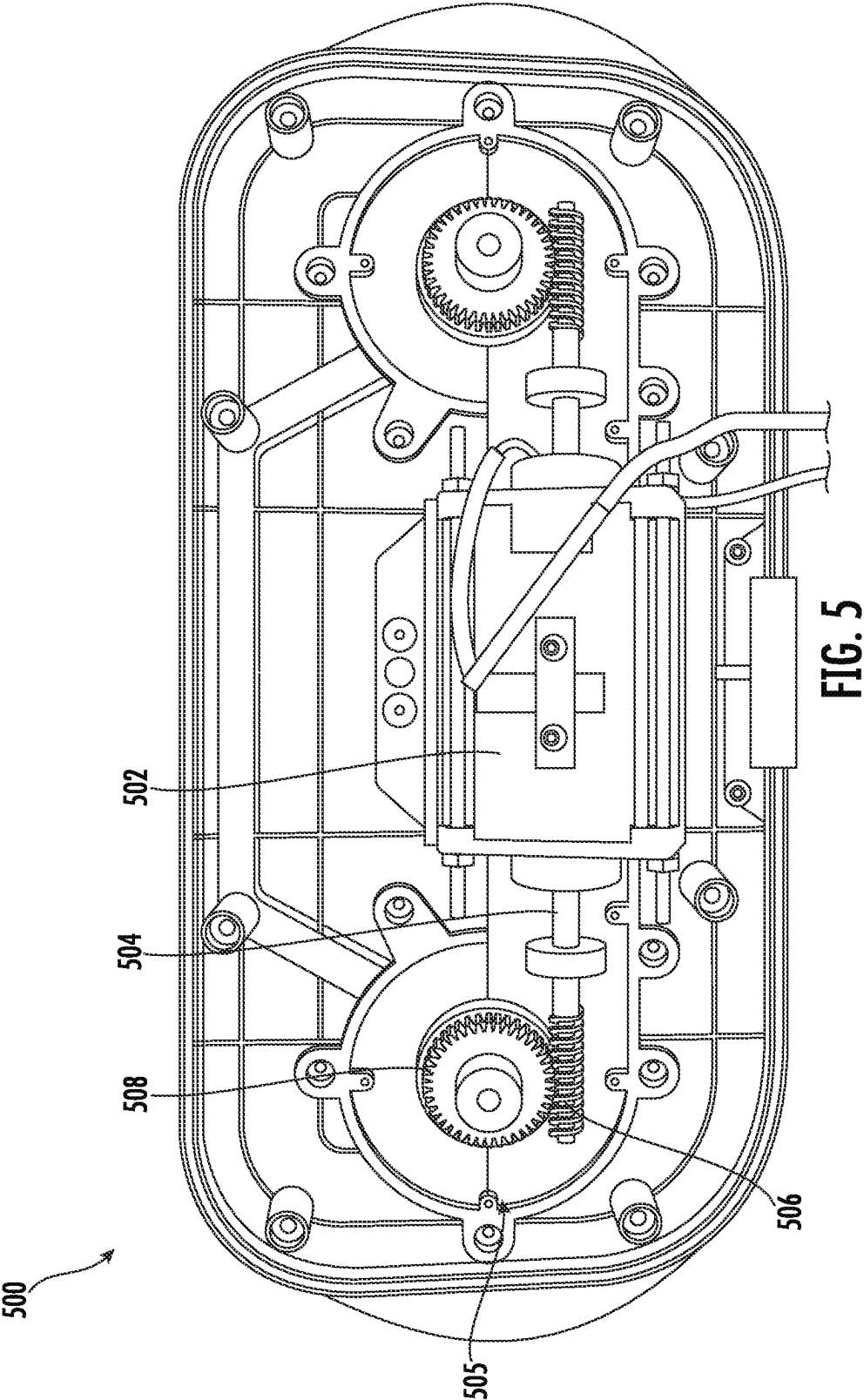


FIG. 4



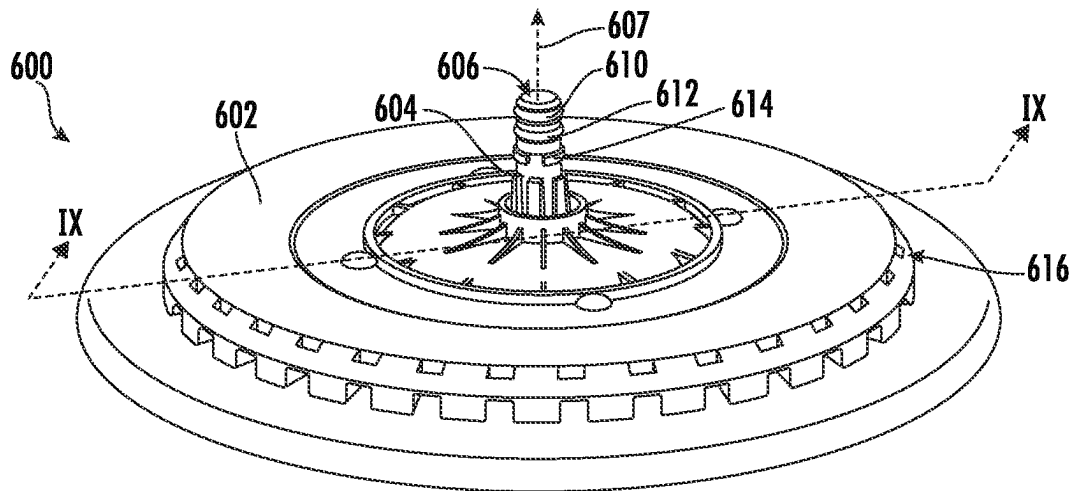


FIG. 6

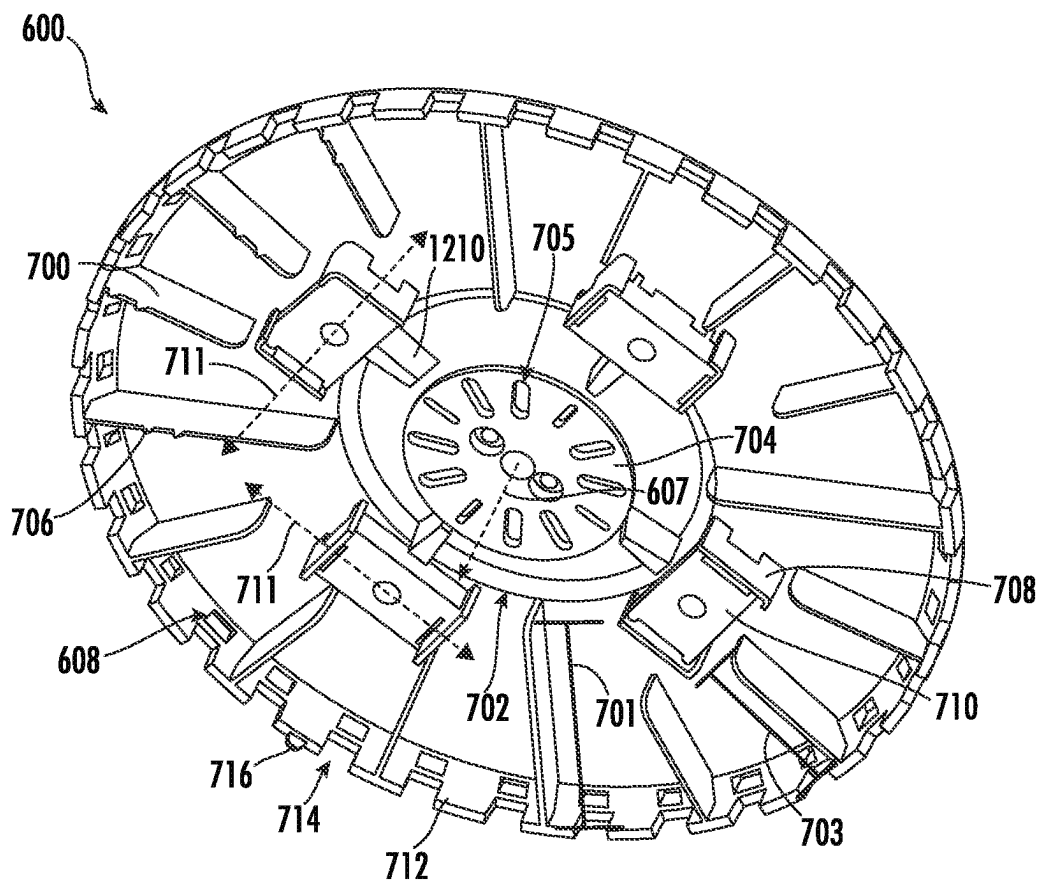


FIG. 7

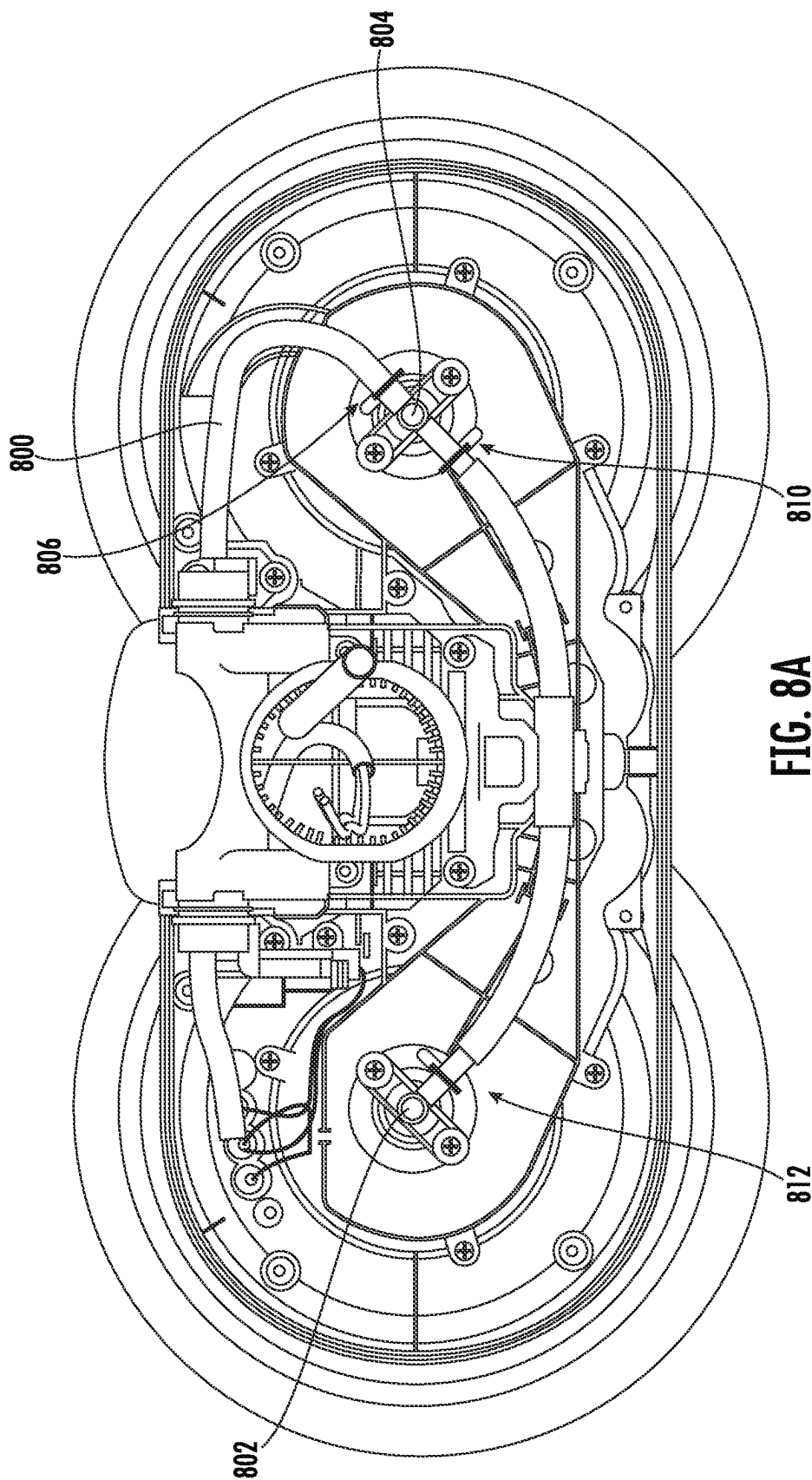


FIG. 8A

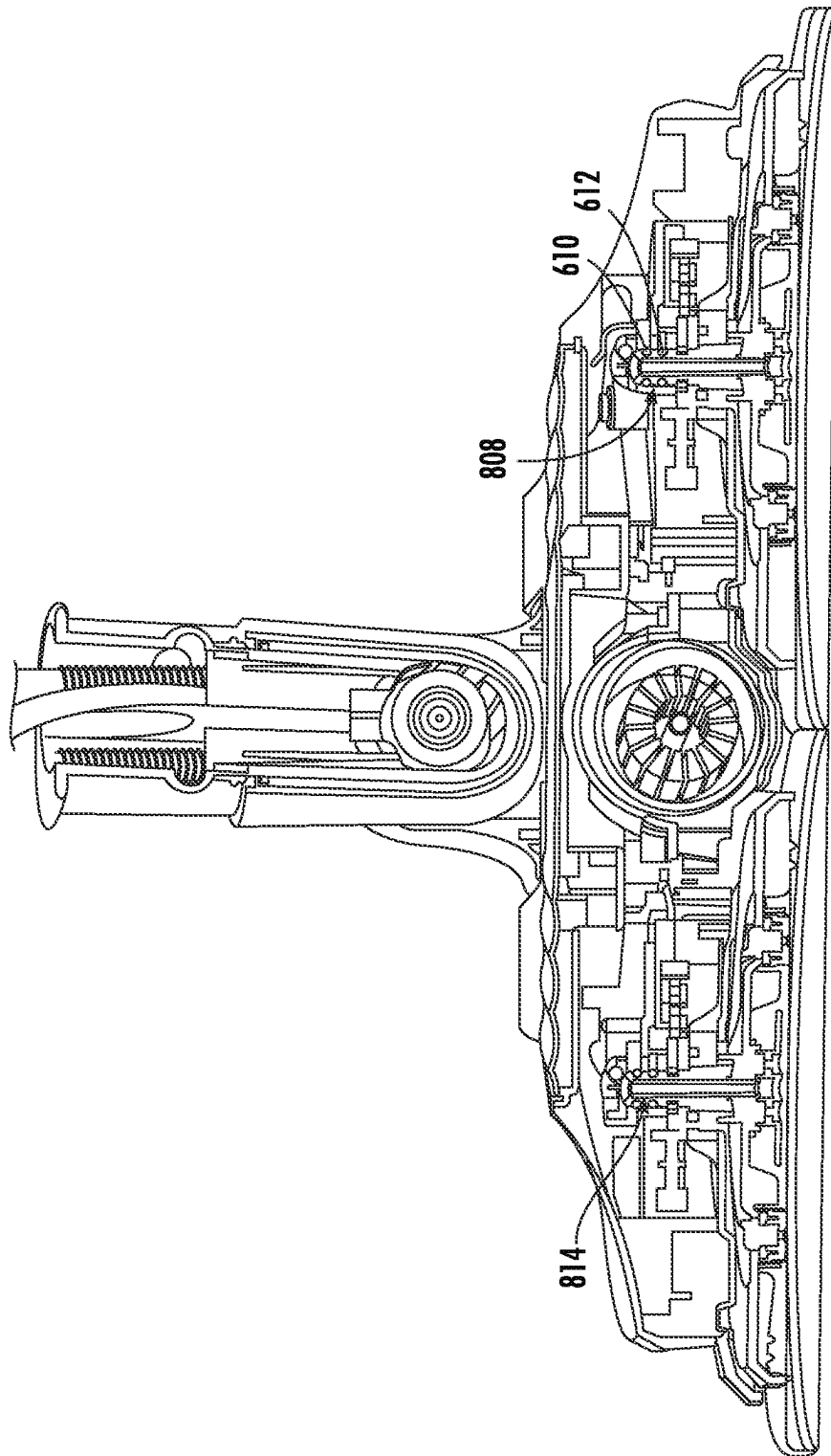


FIG. 8B

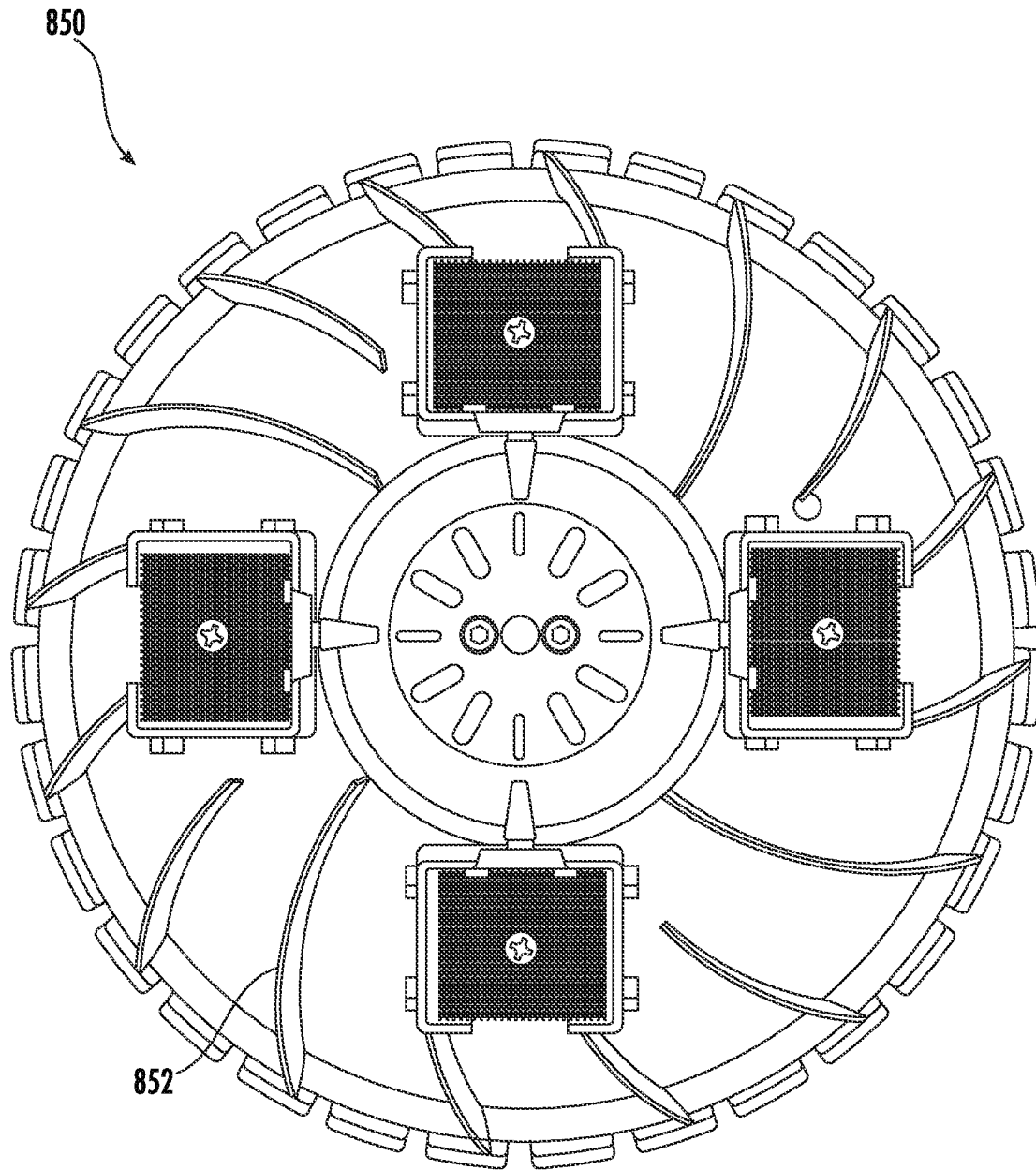


FIG. 8C

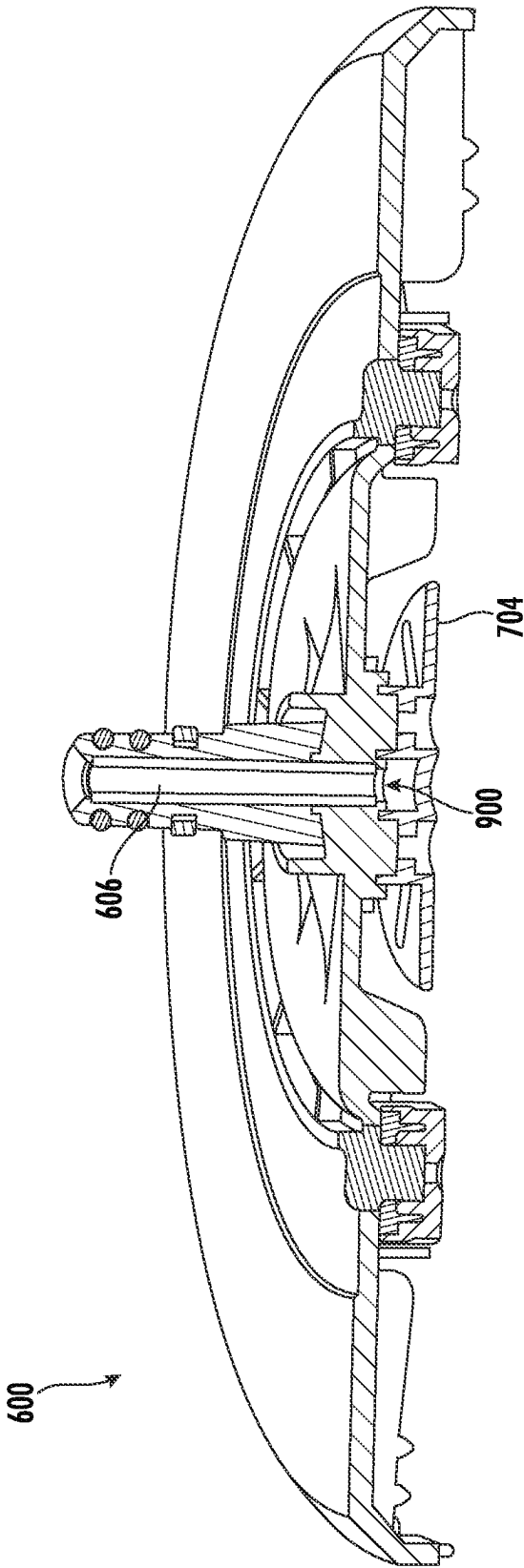


FIG. 9

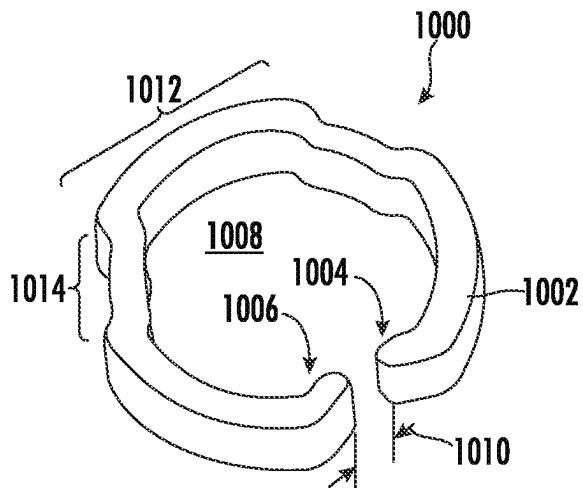


FIG. 10

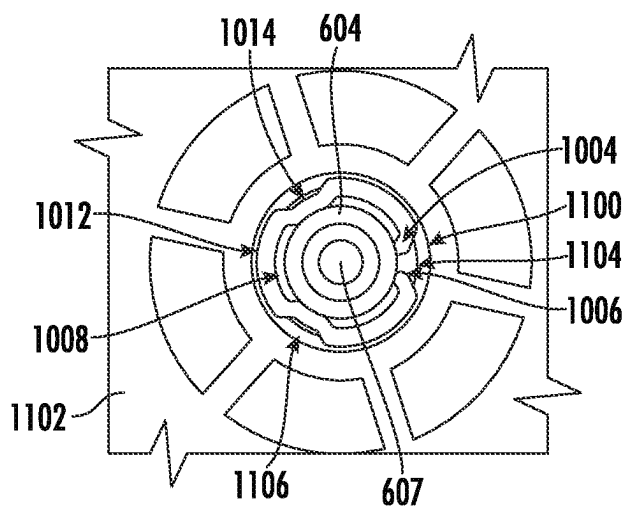
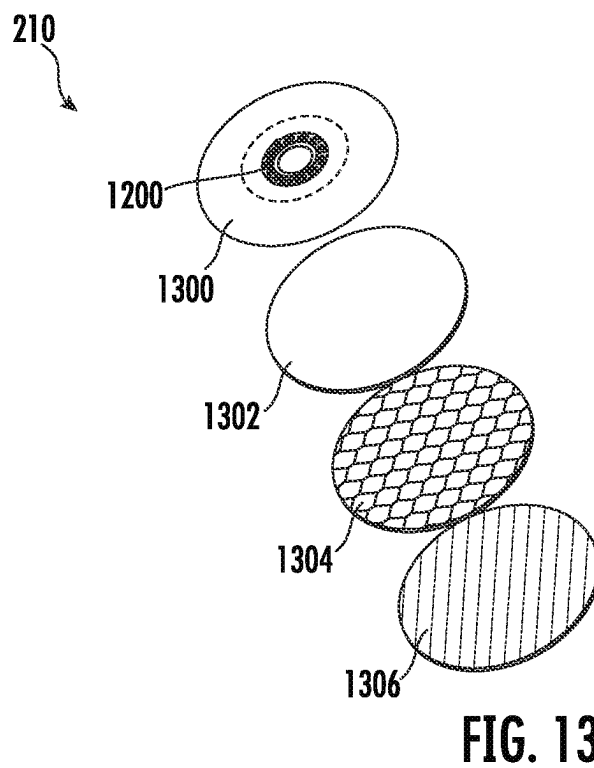
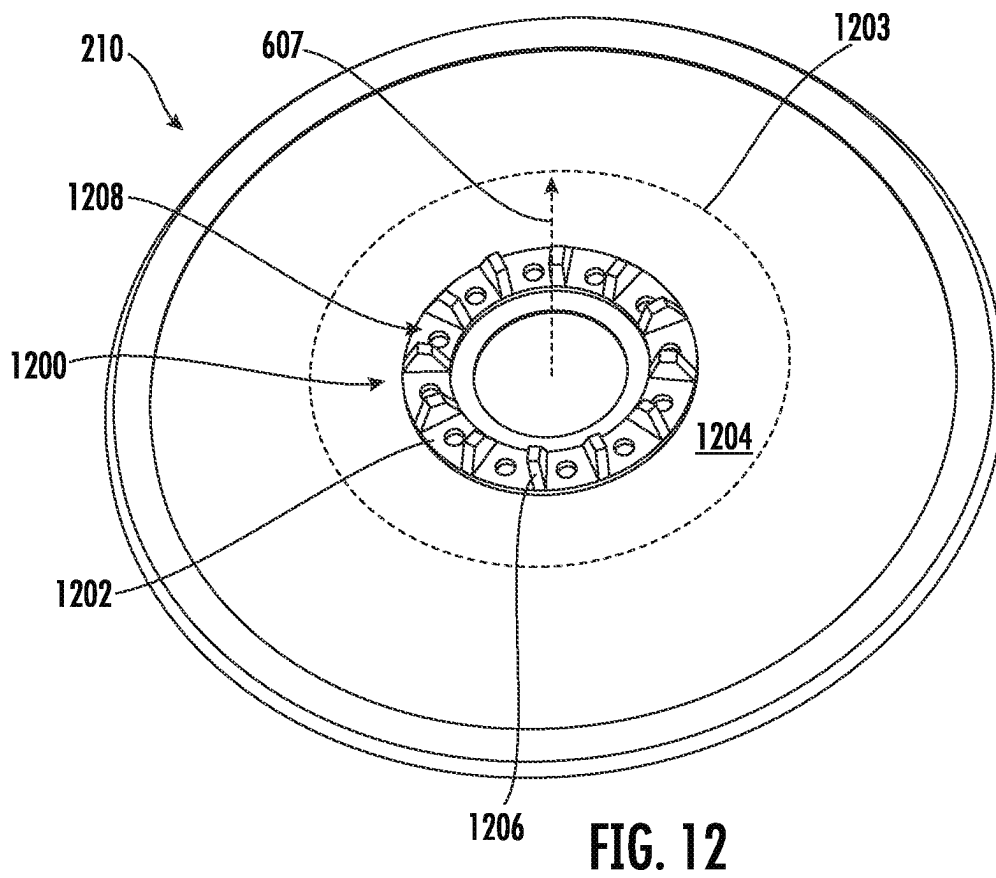


FIG. 11



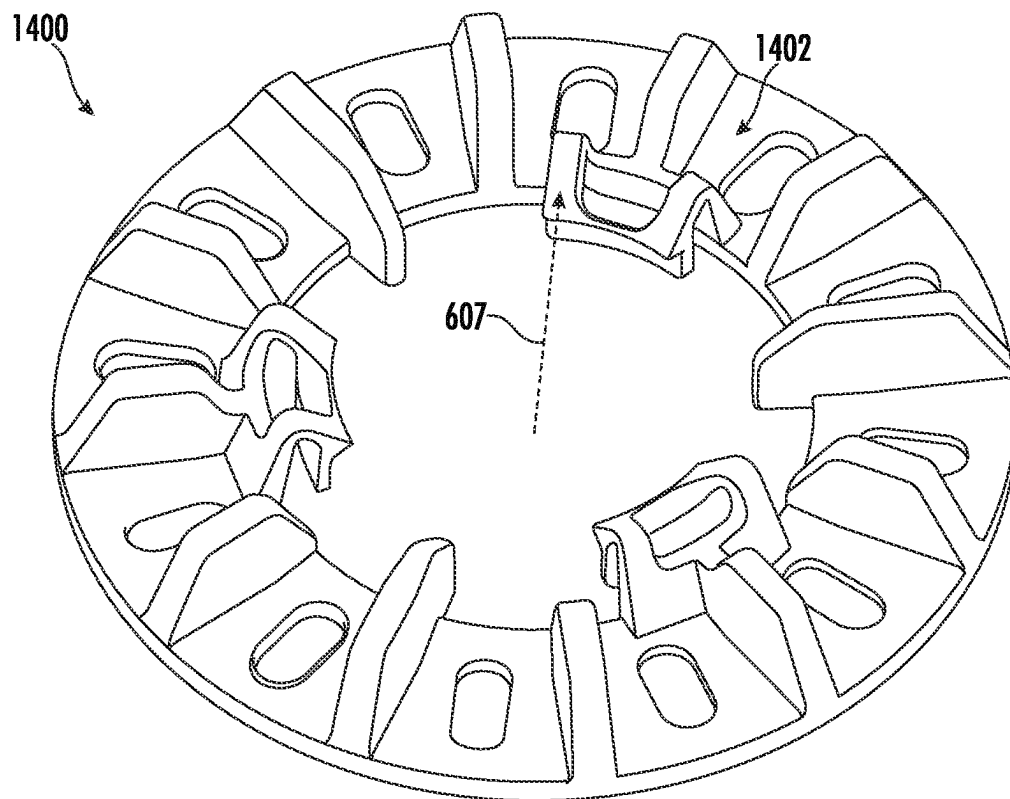


FIG. 14

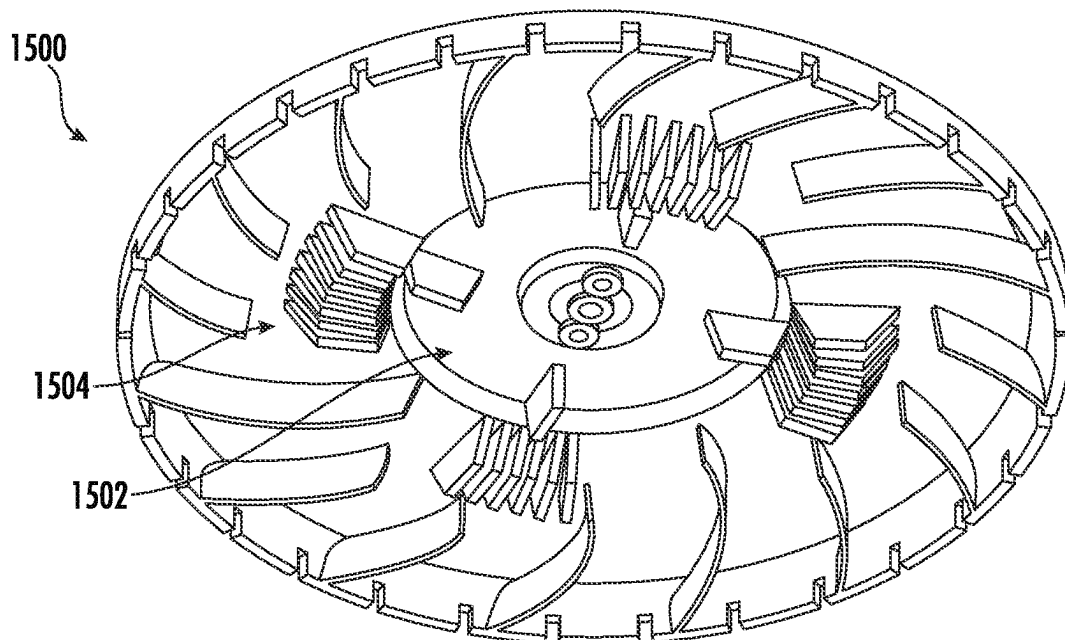


FIG. 15

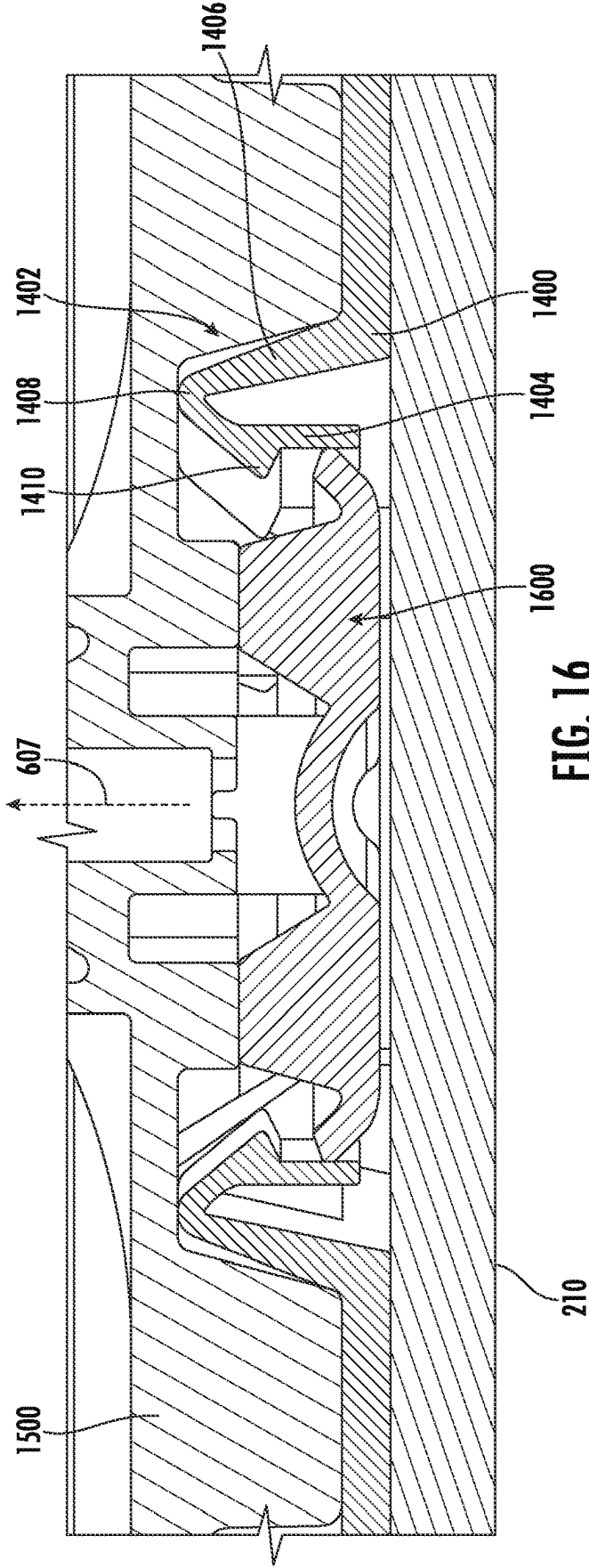


FIG. 16

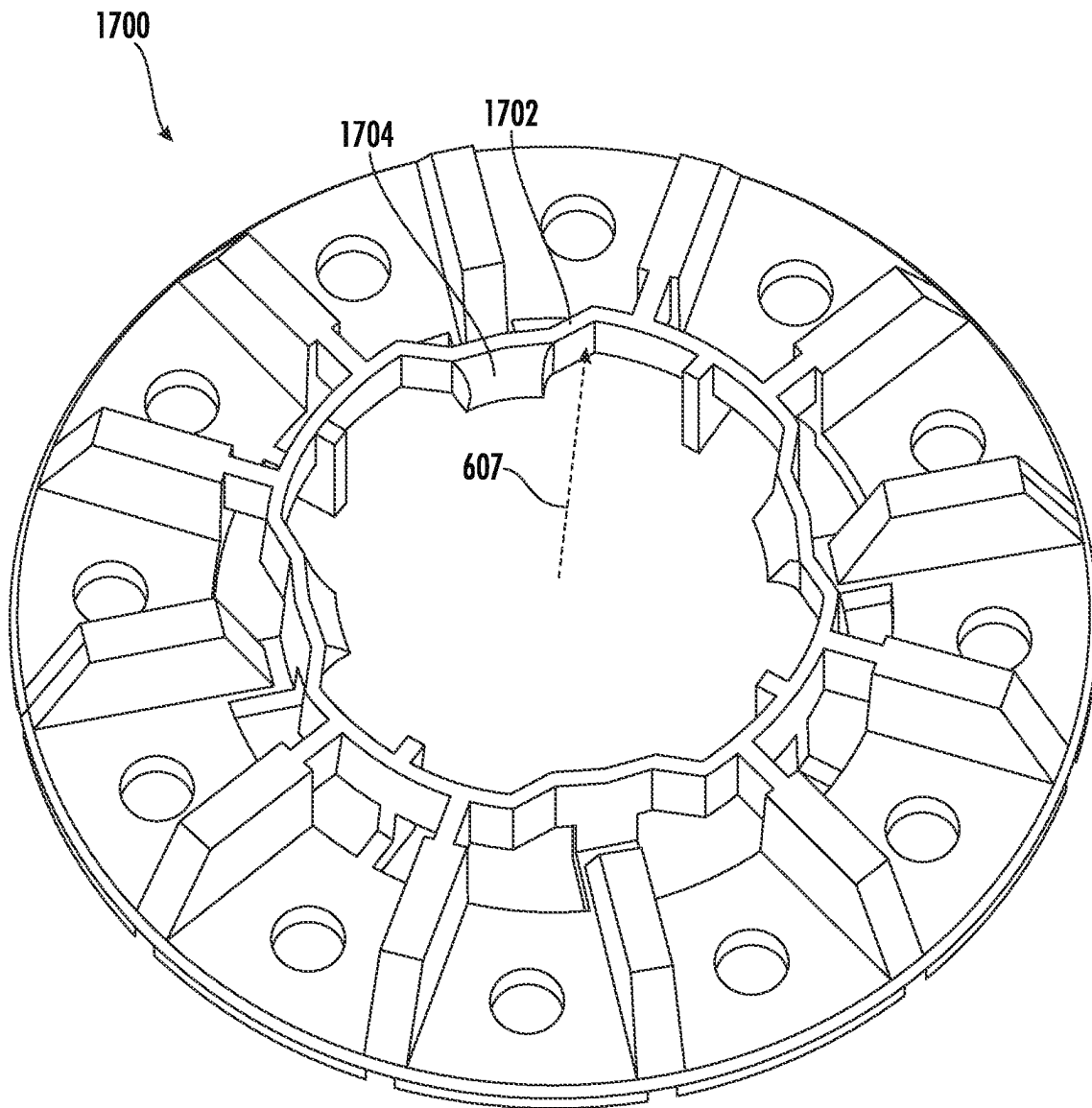


FIG. 17

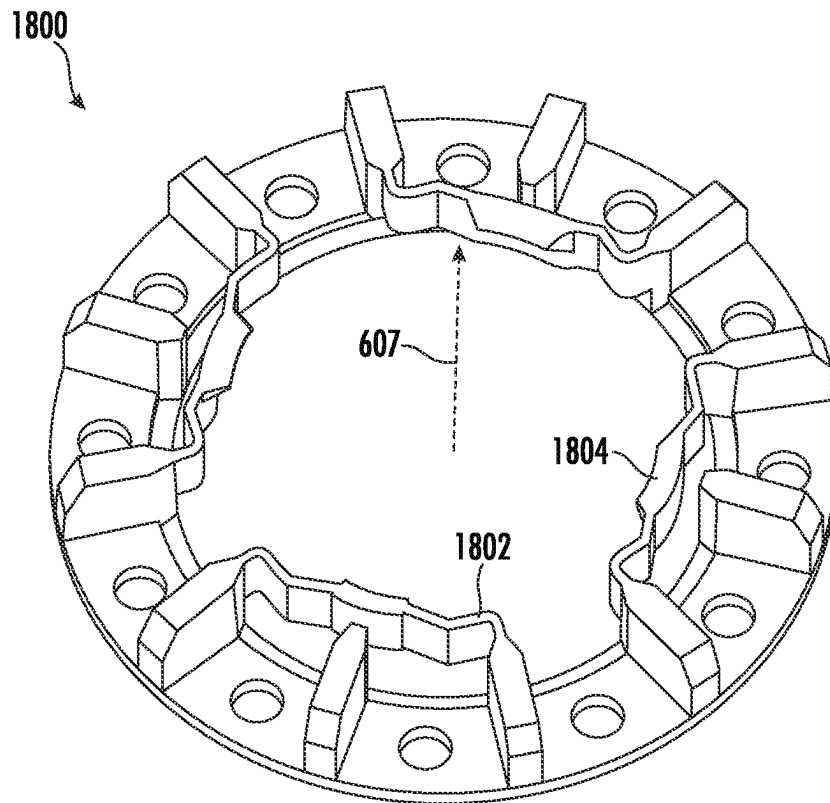


FIG. 18

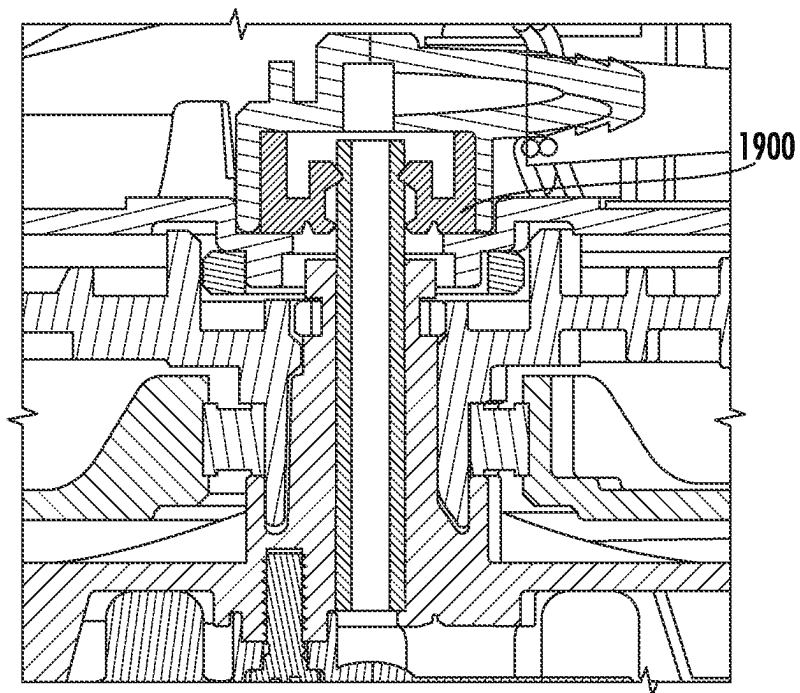
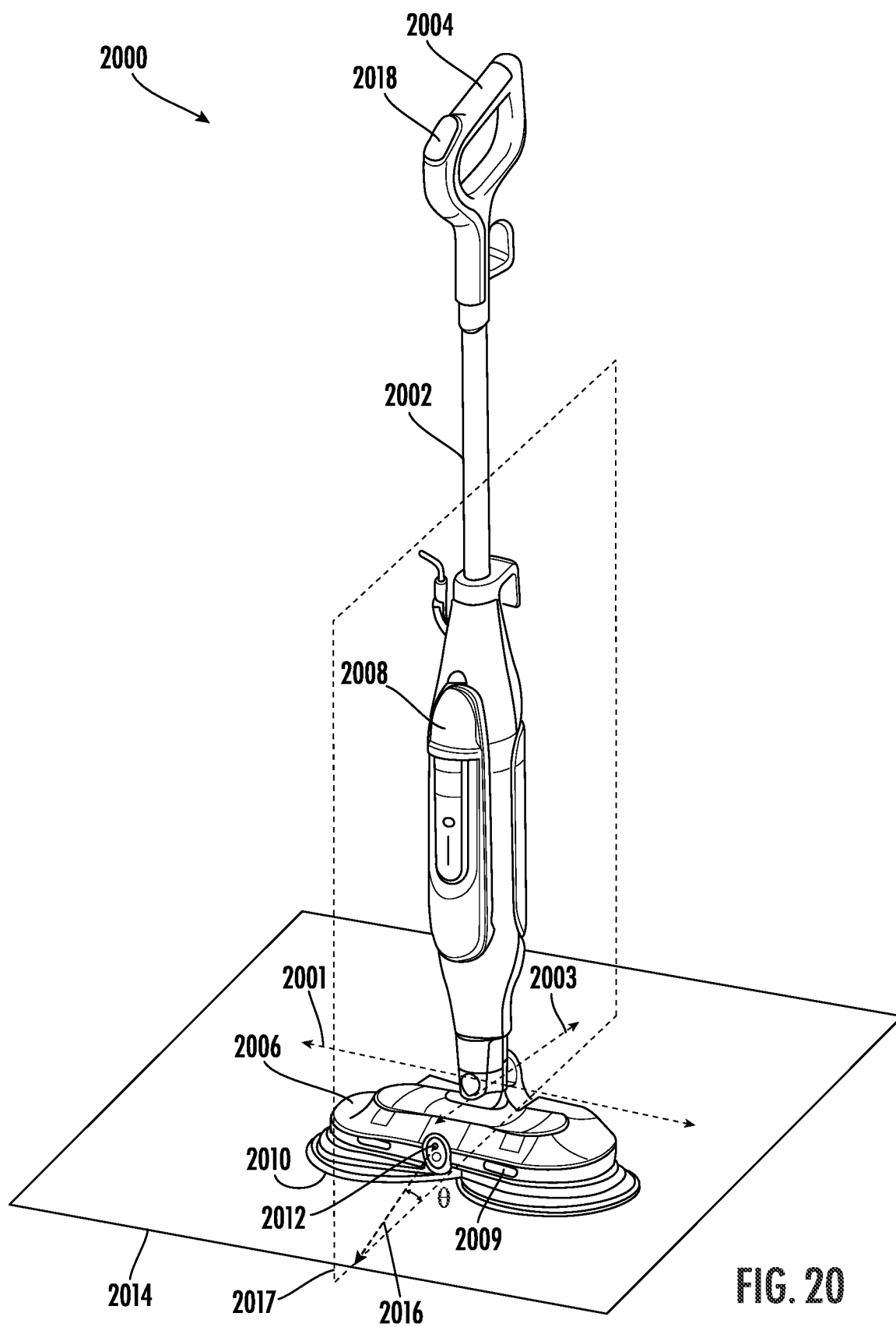
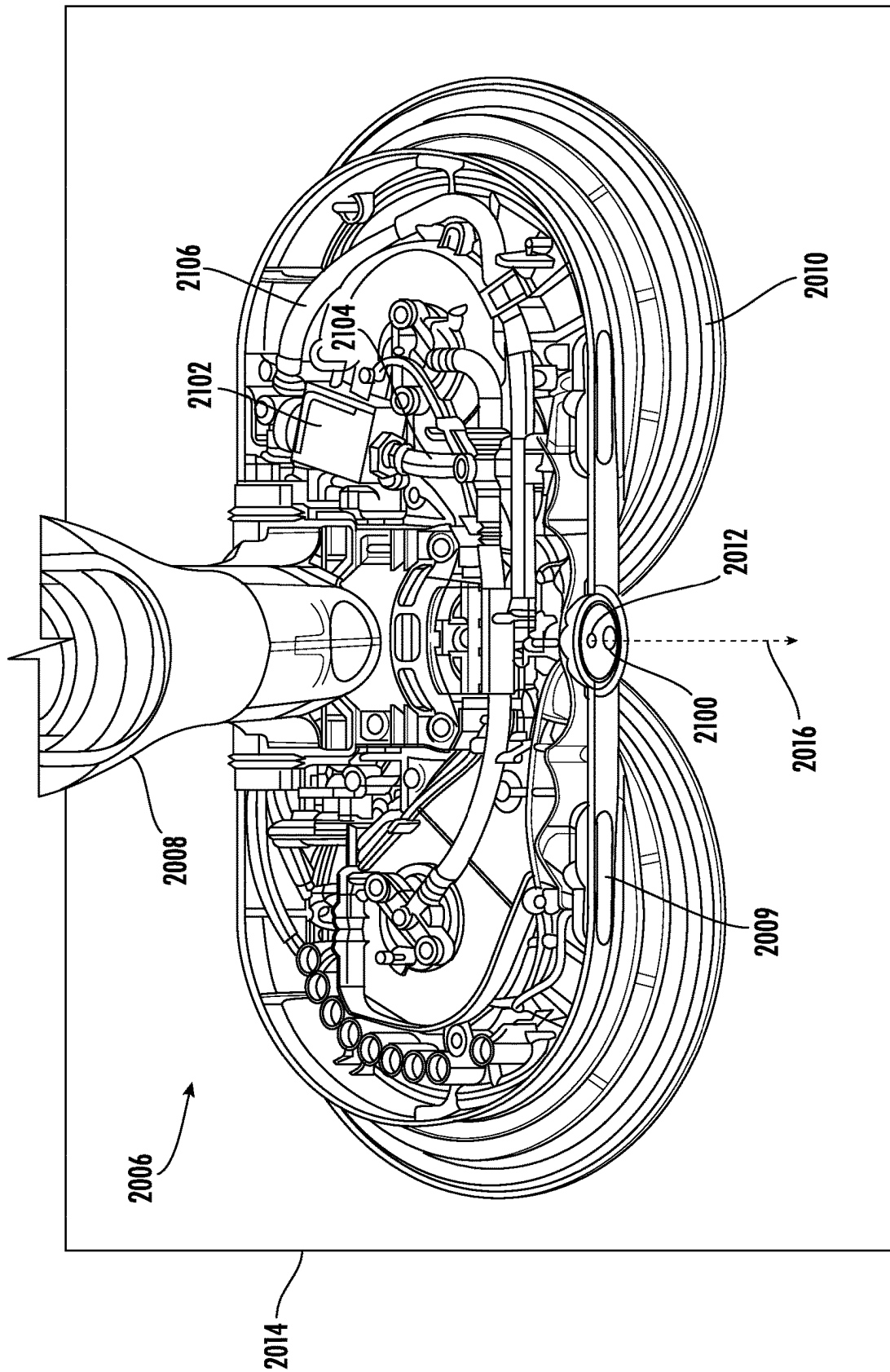


FIG. 19





12345

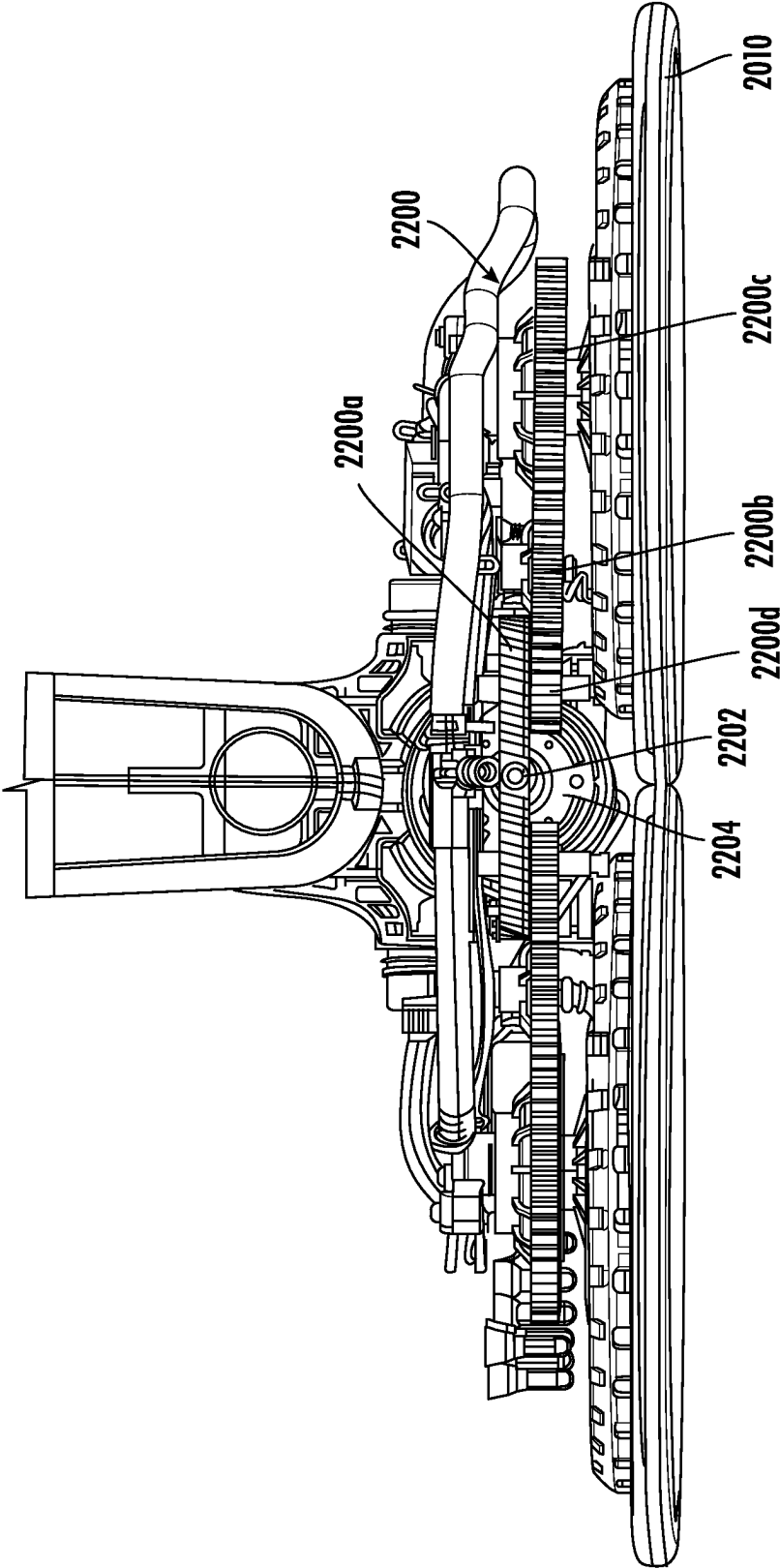


FIG. 22

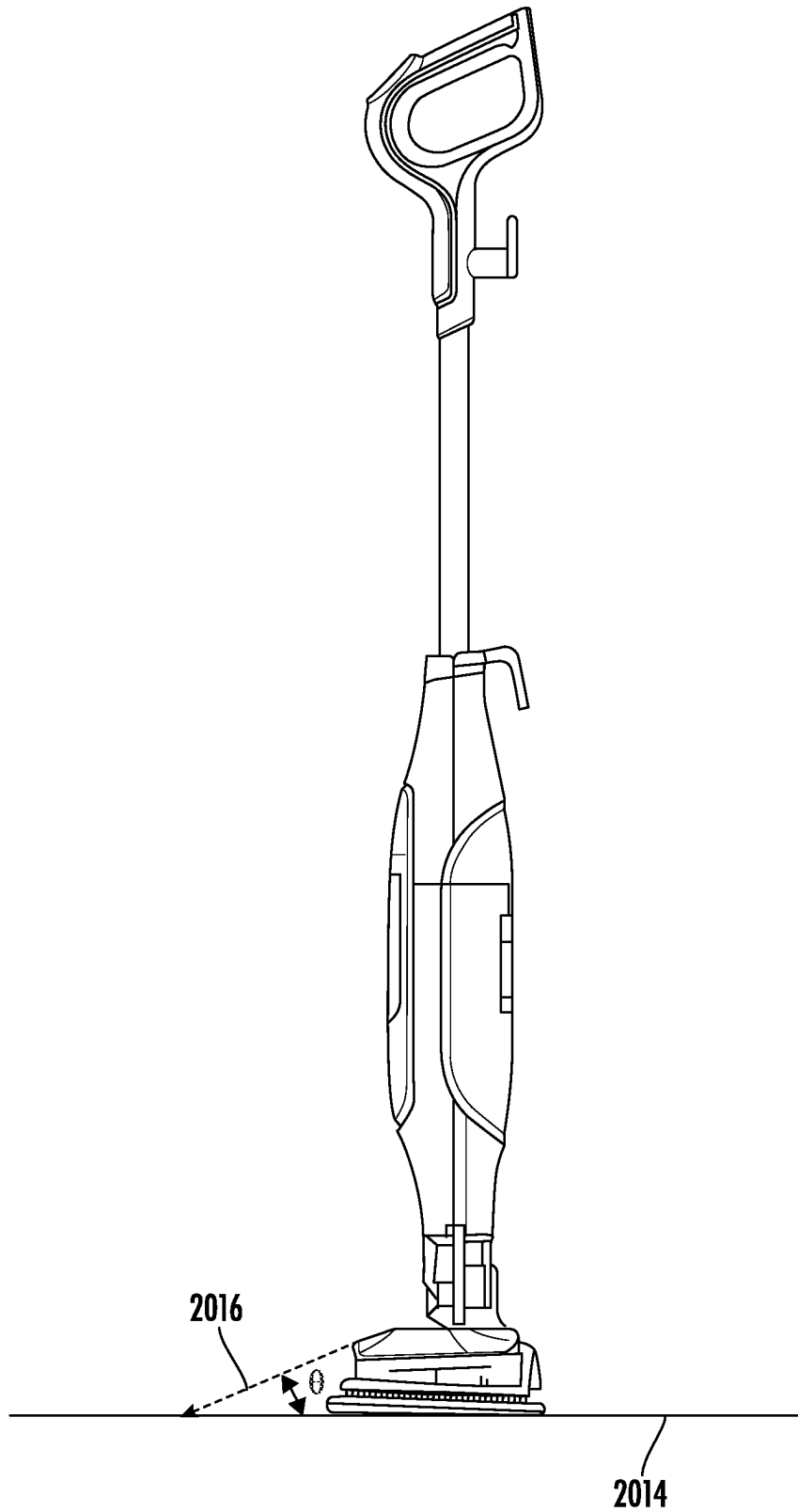


FIG. 23

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STEAM CLEANING APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application claims the benefit of U.S. Provisional Application Ser. No. 63/178,932 filed on Apr. 23, 2021, entitled Steam Cleaning Apparatus and U.S. Provisional Application Ser. No. 63/220,272 filed on Jul. 9, 2021, entitled Steam Cleaning Apparatus, each of which are fully incorporated herein by reference.

TECHNICAL FIELD

The present disclosure generally relates to surface cleaning devices and more specifically to steam cleaning apparatuses.

BACKGROUND INFORMATION

Surface treatment apparatuses can be configured to clean one or more surfaces (e.g., a floor). Surface treatment apparatuses may include, for example, a vacuum cleaner, a mop, a steam cleaning apparatus, a powered broom, and/or any other surface treatment apparatus. A steam cleaning apparatus can include a steam generator, a pad through which steam passes, and a handle for maneuvering the pad along a surface to be cleaned. In some instances, the pad may be agitated (e.g., moved laterally and/or rotationally).

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features and advantages will be better understood by reading the following detailed description, taken together with the drawings wherein:

FIG. 1 shows a schematic example of a steam cleaning apparatus, consistent with embodiments of the present disclosure.

FIG. 2 shows a perspective view of a steam cleaning apparatus, consistent with embodiments of the present disclosure.

FIG. 3 shows a cross-sectional view of an example of a cleaning head of the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 4 shows a cross-sectional perspective view of a portion of another example of a cleaning head of the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 5 shows a cross-sectional view of another example of a cleaning head of the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 6 shows a perspective top view of an example of a pad holder of the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 7 shows a perspective bottom view of the pad holder of FIG. 6, consistent with embodiments of the present disclosure.

FIG. 8A shows a top view an example of a cleaning head of the steam cleaning apparatus of FIG. 2 having a portion a top of the housing removed for purposes of clarity, consistent with embodiments of the present disclosure.

FIG. 8B shows a cross-sectional perspective view of an example of the cleaning head of FIG. 8A, consistent with embodiments of the present disclosure.

FIG. 8C shows a bottom view of another example of a pad holder of the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

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FIG. 9 shows a cross-sectional perspective view of the pad holder of FIG. 6, consistent with embodiments of the present disclosure.

FIG. 10 shows a perspective view of an example of a coupling for coupling a pad holder to the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 11 shows a top view of the coupling of FIG. 10 coupling the pad holder to the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 12 shows a perspective top view of a cleaning pad of the steam cleaning apparatus of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 13 shows an exploded view of the cleaning pad of FIG. 12, consistent with embodiments of the present disclosure.

FIG. 14 shows a perspective view of an example of a pad mount configured to be used with the cleaning pad of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 15 shows a perspective view of an example of a pad holder configured to cooperate with the pad mount of FIG. 14, consistent with embodiments of the present disclosure.

FIG. 16 shows a cross-sectional view of a portion of a pad holder of FIG. 15 having the cleaning pad of FIG. 2 mounted thereto using the pad mount of FIG. 14, consistent with embodiments of the present disclosure.

FIG. 17 shows a perspective view of an example of a pad mount configured to be used with the cleaning pad of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 18 shows a perspective view of an example of a pad mount configured to be used with the cleaning pad of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 19 shows a cross-sectional view of a portion of a cleaning head having an oil seal, consistent with embodiments of the present disclosure.

FIG. 20 shows a perspective view of a steam cleaning apparatus, consistent with embodiments of the present disclosure.

FIG. 21 shows a perspective view of a cleaning head of the steam cleaning apparatus of FIG. 20, wherein a portion of the cleaning head is removed therefrom for purposes of clarity, consistent with embodiments of the present disclosure.

FIG. 22 shows another perspective view of a cleaning head of the steam cleaning apparatus of FIG. 20, wherein a portion of the cleaning head is removed therefrom for purposes of clarity, consistent with embodiments of the present disclosure.

FIG. 23 shows a side view of the steam cleaning apparatus of FIG. 20, consistent with embodiments of the present disclosure.

DETAILED DESCRIPTION

The present disclosure is generally directed to a surface treatment apparatus (e.g., a steam cleaning apparatus). One example of a steam cleaning apparatus may include a cleaning head, a wand having a handle pivotably coupled to the cleaning head, and a steam generator configured to generate steam. The cleaning head includes a plurality of rotating cleaning pads and a pad motor configured to cause each of the rotating cleaning pads to rotate. The cleaning head is fluidly coupled with the steam generator such that steam can pass through each of the plurality of cleaning pads. The pad motor is connected to a drivetrain such that the pad motor can cause each of the cleaning pads to rotate.

In some instances, the drivetrain includes a worm coupled to the pad motor, the worm being configured to engage corresponding gears. In this instance, engagement between the worm and the corresponding gears may result in elevated temperatures within the drivetrain (e.g., due to rotational speed of the worm). As such, one or more temperature sensors may be positioned proximate to the drivetrain (e.g., proximate to the worm). The one or more temperature sensors are configured to monitor a temperature within the drivetrain. In response to the monitored temperature exceeding a threshold, the pad motor may be disabled. Such a configuration may mitigate damage caused to the drivetrain resulting from increased temperatures.

In some instances, the drivetrain includes one or more bevel gears. For example, a motor bevel gear may be coupled to the pad motor and the motor bevel gear may be configured to engage with a drive bevel gear. In this instance, temperature rise may be mitigated when compared to a drivetrain having a worm coupled to the pad motor.

FIG. 1 shows a schematic example of a steam cleaning apparatus 100. As shown, the steam cleaning apparatus 100 includes a wand 102 pivotally coupled to a cleaning head 104. The wand 102 may be pivotally coupled to the cleaning head 104 through a swivel joint. For example, the wand 102 may be pivotable between a storage position (e.g., a vertically extending position) and an in-use position (e.g., a reclined position) about a first pivot axis and may be pivotable between a centered position and one or more side positions (e.g., a left side position and a right side position) about a second pivot axis. One or more components of the steam cleaning apparatus 100 may be electrically coupled to a power source (e.g., a mains power source or a battery power source).

A cleaning assembly 106 is coupled to the wand 102. The cleaning assembly 106 may include, for example, a liquid (e.g., water) reservoir 108, a pump 110 fluidly coupled to the liquid reservoir 108, and a steam generator 112 fluidly coupled to the liquid reservoir 108. In operation, the pump 110 is configured to urge liquid from the liquid reservoir 108 into the steam generator 112. The cleaning assembly 106 is fluidly coupled to the cleaning head 104 such that steam generated by the steam generator 112 is delivered to the cleaning head 104.

The cleaning head 104 includes a plurality of cleaning pads 114 configured to engage with a surface to be cleaned 116 and a pad motor 118 configured to cause each of the cleaning pads 114 to rotate. The cleaning pads 114 may be configured to co-rotate (the cleaning pads 114 each rotate in the same direction) or counter rotate (at least one cleaning pad 114 rotates in a direction different from at least one other cleaning pad 114).

A drivetrain 120 of the cleaning head 104 transfers rotational motion of a motor driveshaft 119 of the pad motor 118 to each of the cleaning pads 114 such that a rotation of the motor driveshaft 119 causes a corresponding rotation in each of the cleaning pads 114. In other words, the drivetrain 120 is configured to transfer rotational motion from the pad motor 118 to each of the cleaning pads 114. As such, a single pad motor 118 can cause both cleaning pads 114 to rotate. In some instances, the drivetrain 120 may generally be referred to as a reduction drivetrain. For example, the drivetrain 120 may be configured to have a 50:1 reduction ratio such that each cleaning pad 114 rotates 50 times slower than the motor driveshaft 119.

The steam delivered to the cleaning head 104 passes through each of the cleaning pads 114 and into contact with

the surface to be cleaned 116. As such, the cleaning pads 114 may generally be described as being fluidly coupled to the cleaning assembly 106.

In some instances, the steam cleaning apparatus 100 may include a controller 122. For example, the controller 122 may be disposed within the cleaning assembly 106, the cleaning head 104, or the wand 102. The controller 122 may be configured to control and/or monitor the steam cleaning apparatus 100. For example, the controller 122 may be configured to receive one or more user inputs corresponding to a cleaning behavior (e.g., pad rotation speed, steam generation rate, and/or any other cleaning behavior). By way of further example, the controller 122 may be configured to monitor a status of the steam cleaning apparatus 100 (e.g., a temperature within the cleaning head, a stall condition of the pad motor 118, and/or any other status).

In some instances the controller 122 may be configured to control one or more features of the steam cleaning apparatus 100 based, at least in part, on a position of the wand 102 relative to the cleaning head 104. For example, when the wand 102 is in the storage position, the pump 110 may be disabled such that liquid is not delivered to the steam generator 112, the pad motor 118 may be disabled such that the cleaning pads 114 no longer rotate, and the steam generator 112 may continue to receive power. When the wand 102 is transitioned to the in-use position, the pump 110 is configured to deliver liquid to the steam generator 112 such that steam is generated and the pad motor 118 is configured to rotate the cleaning pads 114.

FIG. 2 shows a perspective view of an example of a steam cleaning apparatus 200, which may be an example of the steam cleaning apparatus 100 of FIG. 1. As shown, the steam cleaning apparatus 200 includes a wand 202 having a handle 204 pivotally coupled to a cleaning head 206. For example, the wand 202 may be configured to pivot about a first axis 201 between a storage and an in-use position and to pivot about a second axis 203 between a centered position and at least one side position. The first axis 201 may extend transverse (e.g., perpendicular) to a forward direction of movement and the second axis 203 may extend transverse (e.g., perpendicular) to the first axis 201.

A cleaning assembly 208 is coupled to the wand 202 and fluidly coupled to the cleaning head 206. The cleaning assembly 208 is configured to generate steam that flows into the cleaning head 206. As shown, the cleaning head 206 includes a plurality of cleaning pads 210 which are configured to rotate. In operation, steam is configured to pass through the cleaning pads 210.

FIG. 3 shows a cross-sectional view of a cleaning head 300 which may be an example of a cross-section of the cleaning head 206 taken along the line III-III of FIG. 2. As shown, the cleaning head 300 includes a pad motor 302 and a drivetrain 304. The drivetrain 304 includes a worm 306 coupled to a motor driveshaft 303 (shown schematically in hidden lines) of the pad motor 302 and a plurality of gears 308. Rotation of the worm 306 causes a corresponding rotation in the plurality of gears 308. As shown, the plurality of gears 308 include a plurality of worm gears 308a configured to engage with the worm 306, a plurality of intermediary gears 308b, each configured to engage with a respective worm gear 308a, and a plurality of pad gears 308c, each configured to engage with a respective intermediary gear 308b. Each pad gear 308c is coupled to a corresponding pad holder 309 (shown schematically with hidden lines) such that the pad holder 309 rotates with the pad gear 308c. In some instances, the plurality of gears 308 can be configured such that the drivetrain 304 reduces a

rotational speed of the cleaning pads **210** relative to the rotational speed of the worm **306**.

The pad holder **309** is configured to couple to the cleaning pad **210**. In some instances, a pad skirt **311** may extend over at least a portion of the pad holder **309**. The pad skirt **311** may be configured to mitigate or prevent wear on the pad holder **309** resulting from contact between the pad skirt **311** and the pad holder **309**. For example, at least a portion of the pad skirt **311** may include one or more abrasion mitigating members configured to engage at least a portion of the pad holder **309**, wherein the abrasion mitigating members are configured to reduce wear on the pad holder **309**. The one or more abrasion mitigating members may extend along an inner portion of the pad skirt **311**. The one or more abrasion mitigating members may be complaint material (e.g., a rubber) and/or a low friction material (e.g., polytetrafluoroethylene).

Rotation of the worm **306** and the plurality of gears **308** may result in a temperature rise within the cleaning head **300**. Rising temperatures may result in one or more of the plurality of gears **308** failing (e.g., no longer performing as intended). For example, if the plurality of gears **308** are made of a temperature sensitive material (e.g., a plastic), one or more of the plurality of gears **308** may fail when the temperature exceeds a threshold. As such, one or more temperature sensors **310** (shown schematically) may be positioned proximate the drivetrain **304** and be configured to measure a temperature of the drivetrain **304**. For example, one or more temperature sensors **310** may be positioned proximate to one or more of the plurality of gears **308** (e.g., proximate to the worm gear **308a**) and/or the worm **306**. As shown, at least one of the one or more temperature sensors **310** may be positioned above or below the worm **306** (e.g., directly above or below the worm **306** such that a central axis of the temperature sensor **310** is aligned with a central axis of the worm **306**). In response to the one or more temperature sensors **310** sensing a temperature that exceeds a predetermined high temperature threshold, the pad motor **302** may be disabled. The predetermined high temperature threshold may be in a range of, for example, 100 degrees Celsius ($^{\circ}$ C.) and 110 $^{\circ}$ C. By way of further example, the predetermined high temperature threshold may be 105 $^{\circ}$ C. The pad motor **302** may be disabled for a predetermined time and/or until the temperature sensed by the one or more temperature sensors **310** drops below a predetermined operational temperature threshold.

In some instances, the pad motor **302** may include a planetary gear train that is configured to reduce a rotational speed of the motor driveshaft **303** of the pad motor **302**. Such a configuration may mitigate temperature rise caused by rotation of the gears. As such, in these instances, the temperature sensors **310** may be omitted.

FIG. 4 shows a cross-sectional perspective view of a portion of a cleaning head **400**, which may be an example of a cross-section of the cleaning head **206** along the line III-III of FIG. 2. As shown, the cleaning head **400** includes a pad motor **402** and a drivetrain **404**. The drivetrain **404** includes a drive bevel gear **406**, a plurality of step-down bevel assemblies **408**, each having a first and second step-down bevel gear **410** and **412**, and a plurality pad bevel gears **414**. The drive bevel gear **406** is coupled to a motor driveshaft **416** of the pad motor **402** such that the drive bevel gear **406** rotates together with the motor driveshaft **416**.

As shown, the first step-down bevel gear **410** and the second step-down bevel gear **412** are each coupled to a gear driveshaft **418** (e.g., at opposing ends of the gear driveshaft **418**). The gear driveshaft **418** is supported by one or more

bearings **420** (e.g., thrust bearings) such that the gear drive-shaft **418** rotates. The first step-down bevel gear **410** engages the drive bevel gear **406** such that rotation of the drive bevel gear **406** causes a corresponding rotation of the first step-down bevel gear **410**. Rotation of the first step-down bevel gear **410** causes a corresponding rotation in the gear driveshaft **418**. Rotation of the gear driveshaft **418** causes a corresponding rotation of the second step-down bevel gear **412**. The second step-down bevel gear **412** engages the pad bevel gear **414** such that a rotation of the second step-down bevel gear **412** causes a corresponding rotation in the pad bevel gear **414**. The pad bevel gear **414** is coupled to a corresponding pad holder **422** such that a rotation of the pad bevel gear **414** causes a corresponding rotation of the pad holder **422**. The pad holder **422** is configured to couple to the cleaning pad **210**.

As shown, a driveshaft longitudinal axis **424** extends transverse (e.g., perpendicular) to a pad rotation axis **426**. For example, the pad rotation axis **426** may be a vertical axis and the driveshaft longitudinal axis **424** may be a horizontal axis.

Use of bevel gears may mitigate temperature rise caused by rotation of the gears when compared to, for example, the drivetrain **304** of FIG. 3. As such, in these instances, temperature sensors to monitor a temperature of the drivetrain **404** may be omitted.

FIG. 5 shows a cross-sectional view of a cleaning head **500** which may be an example of a cross-section of the cleaning head **206** along the line III-III of FIG. 2. As shown, the cleaning head **500** includes a pad motor **502**. As shown, the pad motor **502** includes a plurality of motor driveshafts **504**, each motor driveshaft **504** extending from an opposing end of the pad motor **502** and configured to engage with a drivetrain **505**. The drivetrain **505** includes a plurality of worms **506** and a plurality of worm gears **508**, each worm **506** is coupled to a respective motor driveshaft **504**. Each worm **506** is configured to engage with a corresponding worm gear **508**. Each worm gear **508** is configured to cause a corresponding pad holder to rotate, wherein each pad holder is configured to couple to a respective cleaning pad **210**.

FIGS. 6 and 7 show a perspective view of a pad holder **600** coupled to the cleaning pad **210** of FIG. 2. The pad holder **600** is configured to be rotationally coupled to the cleaning head **206** of FIG. 2. As shown, the pad holder **600** includes a holder body **602**, a pad driveshaft **604** defining a steam channel **606** therein, and a steam distribution cavity **608** fluidly coupled to the steam channel **606**. The pad driveshaft **604** defines a pad rotation axis **607** that extends transverse to (e.g., perpendicular to) a surface to be cleaned.

The steam channel **606** is fluidly coupled to the cleaning assembly **208** such that steam generated by the cleaning assembly **208** passes through the steam channel **606** and into the steam distribution cavity **608**. The pad driveshaft **604** is configured to couple to a drivetrain of the cleaning head **206** (e.g., the drivetrain **304**, **404**, or **505**) such that the pad holder **600** rotates. As shown, the pad driveshaft further includes a first seal **610**, a second seal **612**, and a coupling **614**. The coupling **614** is configured to couple the pad holder **600** to the drivetrain of the cleaning head **206** (e.g., couples the pad holder **600** to pad gear **308c**, pad bevel gear **414**, or worm gear **508**) such that the pad holder **600** rotates. One example of the coupling **614** may be a C-clip. The seals **610** and **612** are configured to sealingly engage with a steam distribution line **800** (see, FIG. 8A) such that the seals **610** and **612** rotate relative to an inner surface of the steam distribution line. For example, with reference to FIG. 8A,

the steam distribution line **800** includes a terminal steam connector **802** and an intermediary steam connector **804**. The intermediary steam connector **804** has an intermediary steam inlet **806**, a first intermediary steam outlet **808** (see, FIG. **8B**) configured to sealingly engage with the seals **610** and **612** of a respective pad driveshaft **604**, and a second intermediary steam outlet **810** fluidly coupled to the terminal steam connector **802**. The terminal steam connector **802** includes a terminal steam inlet **812** fluidly coupled to the second intermediary steam outlet **810** of the intermediary connector **804** and a terminal steam outlet **814** (see, FIG. **8B**) configured to sealingly engage with the seals **610** and **612** of a respective pad driveshaft **604**. When the seals **610** and **612** are sealingly engaged with the distribution line **800** some steam may escape past the seals **610** and **612**. The seals **610** and **612** may be, for example, O-ring seals. In some instances, the seals **610** and **612** may be replaced with an oil seal **1900** (see, FIG. **19**). The oil seal **1900** may have increased longevity when compared to O-ring seals.

Turning back to FIGS. **6** and **7**, the pad holder **600** may further include one or more holder steam outlets **616**. The one or more holder steam outlets **616** are fluidly coupled to the steam distribution cavity **608** such that steam received by the steam distribution cavity **608** may pass through the one or more holder steam outlets **616**. The one or more holder steam outlets **616** may be configured to divert steam passing therethrough such that the diverted steam does not pass through corresponding cleaning pads **210** (e.g., in a direction away from the cleaning pad **210**). Such a configuration may provide a visual confirmation that steam is being generated.

With reference to FIG. **7**, the steam distribution cavity **608** may include a plurality of ribs **700** extending within the steam distribution cavity **608**, a pad mount receptacle **702**, and a steam diffusion plate **704** configured to diffuse steam passing out of the steam channel **606**. In some instances, the steam diffusion plate **704** may include one or more plate passthroughs **705** through which steam can pass.

As shown, the plurality of ribs **700** extend in a direction that is radially outward from the pad rotation axis **607**. The plurality of ribs **700** are configured to engage (e.g., contact) the cleaning pad **210** such that the ribs **700** support the cleaning pad **210**, encouraging an even force distribution when the cleaning pad **210** engages a surface to be cleaned. The plurality of ribs **700** may include one or more ribs **700** having a first rib length **701** and one or more ribs having a second rib length **703**, the first rib length **701** being greater than the second rib length **703**. The first and second rib lengths **701** and **703** extend in a direction that is radially outward from the pad rotation axis **607**. In some instances, for example, a ratio of a number of ribs **700** having the first rib length **701** to a number of ribs **700** having the second rib length **703** (i.e., a number of ribs **700** having the first rib length **701** divided by a number of ribs having the second rib length **703**) may be 1:3.

In some instances, one or more of the plurality of ribs **700** may include one or more rib protrusions **706** configured to engage (e.g., extend into) the cleaning pad **210** such that movement (e.g., rotational movement) of the cleaning pad **210** relative to the pad holder **600** is resisted by the rib protrusions **706**. For example, and as shown, each rib **700** may include a plurality of rib protrusions **706**, forming a set, wherein a set of rib protrusions **706** corresponding to a first rib **700** are radially offset relative to sets of rib protrusions **706** corresponding to immediately adjacent ribs **700**. In other words, a first plurality of rib protrusions **706** corresponding to a first rib **700** is radially offset from a second plurality of rib protrusions **706** corresponding to immedi-

ately adjacent ribs **700**. By way of further example, a set of rib protrusions **706** corresponding to a first rib **700** may be radially offset relative to sets of rib protrusions **706** corresponding to immediately adjacent ribs **700** such that only a portion of the set of rib protrusions **706** corresponding to the first rib **700** overlaps with only a portion of at least one of the sets of rib protrusions **706** corresponding to immediately adjacent ribs **700**. By way of still further example, a set of rib protrusions **706** corresponding to a first rib **700** may be radially offset relative to sets of rib protrusions **706** corresponding to immediately adjacent ribs **700** such that the set of rib protrusions **706** corresponding to the first rib **700** does not overlap with the sets of rib protrusions **706** corresponding to immediately adjacent ribs **700**. By way of still further example, a set of rib protrusions **706** corresponding to a first rib **700** are not radially offset relative to sets of rib protrusions **706** corresponding to immediately adjacent ribs **700**.

The ribs **700** may further encourage the distribution of steam within the steam distribution cavity **608**. For example, and as shown, adjacent ribs **700** define steam channels that extend radially outward from the pad rotation axis **607**. Steam flowing between adjacent ribs **700** may be encouraged to flow to a perimeter of the steam distribution cavity **608**, encouraging an even distribution of steam within the steam distribution cavity **608**. In some instances, and as shown, the plurality of ribs **700** may be linear. However, the plurality of ribs **700** may have any shape including, for example, an arcuate shape. For example, FIG. **8C** shows a pad holder **850** having arcuate ribs **852**. The arcuate ribs **852** may be curved to correspond to a direction of motion (e.g., such that a convex surface of the arcuate ribs faces the rotation direction or such that a concave surface of the arcuate ribs faces the rotation direction).

As also shown, the steam distribution cavity **608** includes one or more fastener receptacles **708**, each configured to receive a corresponding one or more pad fasteners **710**. The pad fasteners **710** are configured to removably couple the cleaning pad **210** to the pad holder **600**. The pad fasteners **710** are further configured such that movement (e.g., rotational movement) of the cleaning pad **210** relative to the pad holder **600** is resisted by the pad fasteners **710**. The pad fasteners **710** may form a part of a hook and loop fastener, wherein the pad fasteners **710** are one of the hook or the loop and the cleaning pad **210** forms the other one of the hook or the loop. The orientation of the hook and loop fastener may be configured to increase a shear resistance of the connection. For example, the pad fasteners **710** may be oriented such that fastener axes (e.g., longitudinal axes) **711** of at least two immediately adjacent pad fasteners **710** extend in a direction transverse to each other (e.g., at a perpendicular or non-perpendicular angle). Each fastener axis **711** may extend parallel to opposing sides of a respective pad fastener **710**. By way of further example, the pad fasteners **710** may be oriented such that the hooks and/or loops of a hook and loop fastener form a substantially (e.g., within 1°, 2°, 3°, or 4° of) 0°, 45°, or 90° angle with a radial line that extends from the pad rotation axis **607** and intersects the respective hook or loop. Additionally, or alternatively, the hooks and/or loops may be oriented on the pad fastener such that the hooks and/or loops form a substantially 0°, 45°, or 90° angle with a radial line that extends from the pad rotation axis **607** and intersects the respective hook or loop. By way of still further example, the hooks and/or loops may be arranged in a plurality of parallel rows, wherein the hooks and/or loops of a central most row form a substantially 0°, 45°, or 90° with a radial line that extends from the pad rotation axis **607** and intersects each hook and/or loop of the central most row.

The pad fasteners **710** may be, for example, an adhesive tape hook or loop fastener and/or a molded hook or loop fastener. At least a portion of the pad fasteners **710** may be made of, for example, nylon.

The pad holder **600** may further include a plurality of castellations **712** defined in the holder body **602** and positioned along a perimeter of the steam distribution cavity **608**. The castellations **712** may be evenly spaced about the perimeter of the steam distribution cavity **608**. The openings **714** extending between adjacent castellations **712** are fluidly coupled to the steam distribution cavity **608**, allowing steam to pass therethrough. In some instances, the ribs **700** may encourage steam to pass through the openings **714**. One or more of the castellations **712** may include one or more castellation protrusions **716**. The one or more castellation protrusions **716** are configured to engage (e.g., extend into) the cleaning pad **210** such that movement (e.g., rotational movement) of the cleaning pad relative to the pad holder **600** is resisted by the castellation protrusions **716**. In some instances, for example, there may be at least four castellation protrusions **716**, wherein the castellation protrusions **716** are evenly spaced about a perimeter of the steam distribution cavity **608**.

FIG. **9** shows a cross-sectional perspective view of the pad holder **600** taken along the line IX-IX of FIG. **6**. As shown, the steam diffusion plate **704** is proximate a steam outlet **900** of the steam channel **606** such that at least a portion of the steam passing through the steam outlet **900** is incident of the steam diffusion plate **704**.

FIG. **10** shows a perspective view of an example of a coupling **1000** that is configured to couple the pad holder **600** to the drivetrain of the cleaning head **206** (e.g., couples the pad holder **600** to pad gear **308c**, pad bevel gear **414**, or worm gear **508**). The coupling **1000** may be an example of the coupling **614** of FIG. **6**.

As shown, the coupling **1000** includes a coupling body **1002** having a first end **1004** and a second end **1006** that partially encloses an area **1008** configured for receiving a portion of the pad driveshaft **604**. The first and second ends **1004** and **1006** are spaced apart from each other by a separation distance **1010** such that the coupling body **1002** partially encloses the area **1008**. As shown, the coupling body **1002** defines a plurality of arcuate regions **1012**, wherein immediately adjacent arcuate regions **1012** are each separated by a recessed region **1014**. The recessed region **1014** may be a non-arcuate region (e.g., a planar region). The coupling body **1002** may be formed of glass filled nylon (e.g., nylon having a 15% glass fill).

As shown in FIG. **11**, the arcuate regions **1012** engage with corresponding arcuate regions **1100** defined by a gear **1102** of the drivetrain of the cleaning head **206**. The arcuate regions **1100** defined by the gear **1102** may define at least a portion of a receptacle **1104** for receiving the coupling **1000**. As shown, the receptacle **1104** may include a base **1106** configured to support the coupling **1000**.

As also shown, the coupling body **1002** extends around the pad rotation axis **607**. The coupling body **1002** can be configured to engage (e.g., contact) the pad driveshaft **604**. For example, the coupling body **1002** can contact the pad driveshaft **604** at the recessed regions **1014** and at the first and second ends **1004** and **1006** and the coupling body **1002** can be spaced apart from the pad driveshaft **604** at the arcuate regions **1012**. As such, the coupling **1000** may generally be described as having at least four points of contact with the pad driveshaft **604**. Engagement between the pad driveshaft **604** and the coupling body **1002** may align the coupling body **1002** relative to the pad driveshaft

604 such that, for example, the pad rotation axis **607** extends centrally through the area **1008** partially enclosed by the coupling body **1002**. In some instances, engagement between the pad driveshaft **604** and the coupling body **1002** may result in improved retention of the pad driveshaft **604** (e.g., when compared to a C-clip).

FIG. **12** shows a perspective view of the cleaning pad **210**. As shown, the cleaning pad **210** includes a pad mount **1200** configured to cooperate with the pad mount receptacle **702**. The pad mount receptacle **702** may define a recessed region configured to receive at least a portion of the pad mount **1200**. The pad mount **1200** may include a base **1202** (e.g., an annular base) that extends around (e.g., centrally around) the pad rotation axis **607**. The annular base **1202** is coupled to a mounting surface **1204** of the cleaning pad **210**. For example, the annular base **1202** may be coupled to the mounting surface **1204** using any one or more of an adhesive, stitching, and/or any other form of coupling. At least a portion of the mounting surface **1204** may form part of a hook and loop fastener. For example, the mounting surface **1204** may include an annular fastener region **1203** extending around the annular base **1202**. The annular fastener region **1203** may form part of a hook and loop fastener.

The pad mount **1200** may further include one or more mount protrusions **1206** that extend from the annular base **1202**. For example, a plurality of mount protrusions **1206** may extend along the annular base **1202** and may be spaced apart from each other in regular intervals. In this example, one or more of the mount spaces **1208** defined between immediately adjacent mount protrusions **1206** may be configured to receive a receptacle protrusion **1210** (see, FIG. **7**) extending within the pad mount receptacle **702**. As such, the pad mount **1200** may be configured to resist movement (e.g., rotational movement) of the cleaning pad **210** relative to the pad holder **600**. As shown, there may be fewer receptacle protrusions **1210** than mount spaces **1208** defined between immediately adjacent mount protrusions **1206**, such a configuration may encourage easier alignment of the pad mount **1200** with the pad mount receptacle **702**.

As shown, the one or more mount protrusions **1206** may have a generally triangular shape with an arcuate (or rounded) vertex. However, the one or more mount protrusions **1206** may have other shapes (e.g., cylindrical, rectangular, or any other shape). The one or more mount protrusions **1206** may be configured to encourage alignment of the cleaning pad **210** with the pad holder **600**.

FIG. **13** shows an exploded view of the cleaning pad **210**. As shown, the cleaning pad **210** includes a plurality of layers. For example, the cleaning pad **210** may include a mounting layer **1300**, a stiffening layer **1302**, a netting layer **1304**, and a cleaning layer **1306**. The pad mount **1200** is coupled to the mounting layer **1300** and the mounting layer **1300** may include a material configured to couple to the pad fasteners **710**. For example, the mounting layer **1300** may include a material that forms a hook or loop portion of a hook and loop fastener. The stiffening layer **1302** may be configured to stiffen the cleaning pad **210**. Stiffening the cleaning pad **210** may mitigate a wrinkling of the cleaning pad **210** when rotated against a surface to be cleaned. The stiffening layer **1302** may be formed of a polyethylene mesh material (although other materials may be used). The netting layer **1304** may be a mesh material having a pore size that is greater than the pore size of the mesh material forming the stiffening layer **1302**. The cleaning layer **1306** is configured to engage a surface to be cleaned. The cleaning layer **1306** may be, for example, a microfiber material (e.g., including polyester, olefin, and/or nylon). As shown, the stiffening

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layer **1302** is disposed between the netting layer **1304** and the mounting layer **1300** and the netting layer is disposed between the stiffening layer **1302** and the cleaning layer **1306**. As such, the stiffening layer **1302** and the netting layer **1304** may generally be described as being internal layers. In operation, steam is capable of passing through each of the mounting layer **1300**, the stiffening layer **1302**, the netting layer **1304**, and the cleaning layer **1306**.

FIG. **14** shows an example of a pad mount **1400** configured to be used with the cleaning pad **210** and FIG. **15** shows an example of a pad holder **1500** configured to cooperate with the pad mount **1400** and FIG. **16** shows a cross-sectional view of a portion of a pad holder **1500** having the cleaning pad **210** mounted thereto using the pad mount **1400**. As shown, the pad mount **1400** includes snap arms **1402** configured to releasably couple to a steam diffusion plate **1600** of the pad holder **1500**. For example, the snap arms **1402** can be configured to form a snap fit connection with the steam diffusion plate **1600**. The snap arms **1402** can extend in a direction of the pad rotation axis **607**. For example, and as shown, the snap arms **1402** can generally be described as having a U-shape having a first snap leg **1404**, a second snap leg **1406**, and a connecting region **1408** connecting the first and second snap legs **1404** and **1406**. The first snap leg includes a connector **1410** for coupling to the steam diffusion plate **1600**. In operation, when a user couples the pad mount **1400**, the steam diffusion plate **1600** urges the first snap leg **1404** towards the second snap leg **1406**. Once connected, the first snap leg **1404** moves back towards a rest position and the connector **1410** couples the pad mount **1400** to the steam diffusion plate **1600**.

The pad holder **1500** may include a pad mount receptacle **1502** and one or more pad alignment protrusions **1504** arranged about the pad mount receptacle **1502**. The pad mount receptacle **1502** is configured to receive at least a portion of the pad mount **1400**. The pad alignment protrusions **1504** are configured to encourage alignment of the pad mount **1400** when being coupled to the pad holder **1500**.

FIG. **17** shows another example of a pad mount **1700** configured to form a snap-fit with the steam diffusion plate **1600**. As shown, the pad mount **1700** includes a snap arm **1702** having a centrally located connector **1704**. In operation, when a user couples the pad mount **1700** to the steam diffusion plate **1600**, at least a portion of the central portion of the snap arm **1702** is urged in a direction away from the pad rotation axis **607**. Once connected, the central portion of the snap arm **1702** moves back towards a rest position and the connector **1704** couples the pad mount **1700** to the steam diffusion plate **1600**.

FIG. **18** shows another example of a pad mount **1800** configured to form a snap-fit with the steam diffusion plate **1600**. As shown, the pad mount **1800** includes a snap arm **1802** having a centrally located connector **1804**. In operation, when a user couples the pad mount **1800** to the steam diffusion plate **1600**, at least a portion of the central portion of the snap arm **1802** is urged in a direction away from the pad rotation axis **607**. Once connected, the central portion of the snap arm **1802** moves back towards a rest position and the connector **1804** couples the pad mount **1800** to the steam diffusion plate **1600**.

FIG. **20** shows a perspective view of an example of a steam cleaning apparatus **2000**, which may be an example of the steam cleaning apparatus **100** of FIG. **1**. As shown, the steam cleaning apparatus **2000** includes a wand **2002** having a handle **2004** pivotally coupled to a cleaning head **2006**. For example, the wand **2002** may be configured to pivot about a first axis **2001** between a storage and an in-use position and

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to pivot about a second axis **2003** between a centered position and at least one side position. The first axis **2001** may extend transverse (e.g., perpendicular) to a forward direction of movement and the second axis **2003** may extend transverse (e.g., perpendicular) to the first axis **2001**.

A cleaning assembly **2008** is coupled to the wand **2002** and fluidly coupled to the cleaning head **2006**. The cleaning assembly **2008** is configured to generate steam that flows into the cleaning head **2006**. As shown, the cleaning head **2006** includes a plurality of cleaning pads **2010** which are configured to rotate. In operation, steam is configured to pass through the cleaning pads **2010**. As shown, the cleaning assembly **2008** may include one or more environment illuminating elements **2009** (e.g., one or more incandescent bulbs, one or more light emitting diodes, and/or any other lighting element) configured to illuminate at least a portion of a surrounding environment.

The cleaning head **2006** may further include a steam nozzle **2012**. The steam nozzle **2012** is fluidly coupled to the cleaning assembly **2008** such that steam generated by the cleaning assembly **2008** passes through the steam nozzle **2012**. The steam nozzle **2012** is configured to direct steam in a direction of a surface to be cleaned **2014** (e.g., a floor) along a steam axis **2016**. The steam axis **2016** can be configured to intersect the surface to be cleaned **2014** at a location in front of the cleaning head **2006**. Such a configuration may allow a user to apply steam to a location in the front of the cleaning head **2006** prior to positioning the cleaning pads **2010** at the location for scrubbing. The additional steam from the steam nozzle **2012** may soften debris adhered to the surface to be cleaned **2014**. For example, and as shown, the steam axis **2016** may extend within a vertical plane **2017** that is disposed between the cleaning pads **2010** and intersects the cleaning assembly **2008**. In some instances, the second axis **2003** may extend within the vertical plane **2017**.

The steam axis **2016** may form an intersection angle θ with the surface to be cleaned **2014** (see also, FIG. **23**). The intersection angle θ may be in a range of, for example, 1° to 89° . By way of further example, the intersection angle θ may be in a range of 15° to 75° . By way of still further example, the intersection angle θ may be in a range of 20° to 70° . By way of still further example, the intersection angle θ may be in a range of 25° to 65° . By way of still further example, the intersection angle θ may be in a range of 30° to 60° . By way of still further example, the intersection angle θ may be in a range of 10° to 80° .

The handle **2004** can include a user interface **2018** (e.g., one or more buttons) configured to control one or more behaviors of the steam cleaning apparatus **2000**. For example, the user interface **2018** may be configured to control an agitation speed (e.g., a rotation speed) of the cleaning pads **2010**. By way of further example, the user interface **2018** may be configured to control a quantity of steam generated that passes through one or more of the cleaning pads **2010** and/or the steam nozzle **2012**. By way of still further example, the user interface **2018** may be configured to control whether steam passes through only the cleaning pads **2010**, only the steam nozzle **2012**, and/or both the cleaning pads **2010** and the steam nozzle **2012**.

As shown in FIG. **21**, in some instances, the cleaning head **2006** may include one or more steam illuminating elements **2100** (e.g., one or more incandescent bulbs, one or more light emitting diodes, and/or any other lighting element). The one or more steam illuminating elements **2100** may be configured to illuminate a region of the surface to be cleaned **2014** within which the steam axis **2016** intersects the surface

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to be cleaned **2014**. In some instances, the one or more steam illuminating elements **2100** may be configured to illuminate the surface to be cleaned **2014** before steam passes through the steam nozzle **2012**, allowing a user to more readily identify the region of the surface to be cleaned **2014** where steam will be applied. Additionally, or alternatively, the one or more steam illuminating elements **2100** may be configured to illuminate the steam passing through the steam nozzle **2012**. Such a configuration may allow a user to more readily confirm that steam is passing through the steam nozzle **2012**.

As also shown in FIG. **21**, the cleaning head **2006** includes a steam valve **2102**. The steam valve **2102** is fluidly coupled to the cleaning assembly **2008**. The steam valve **2102** is configured to selectively direct steam passing through to one or both of the steam nozzle **2012** and/or each of the cleaning pads **2010**. For example, the steam valve **2102** may be configured such that steam generated by the cleaning assembly **2008** is selectively delivered to only one of the steam nozzle **2012** or the cleaning pads **2010** as a time. The steam valve **2102** may include an off-position in which steam is substantially prevented from flowing through the steam valve **2102**. The steam valve **2102** may be responsive to inputs received from the user interface **2018**. For example, the steam valve **2102** may be an electronic valve such as a solenoid valve or motorized valve.

As shown, the steam valve **2102** is fluidly coupled a pad distribution line **2104** and a nozzle distribution line **2106**. The pad distribution line **2104** fluidly couples the steam valve **2102** to the cleaning pads **2010** such that steam passes through the cleaning pads **2010**. The nozzle distribution line **2106** fluidly couples the steam valve **2102** to the steam nozzle **2012**. The distribution lines **2104** and **2106** may be an elastomeric (e.g., natural or synthetic rubber) tubing, a metal tubing, a plastic tubing, and/or any other tubing.

As shown in FIG. **22**, each of the cleaning pads **2010** are configured to rotate in response to a rotation of a plurality of gears **2200**. As shown, the plurality of gears **2200** includes a plurality of worm gears **2200a**, a plurality of intermediary gears **2200b**, a plurality of pad gears **2200c**, and a plurality of drive gears **2200d**. A worm **2202** is configured to engage each of the worm gears **2200a** such that each of the worm gears **2200a** rotate in response to rotation of the worm **2202**. The worm **2202** is coupled to a driveshaft of a pad motor **2204**. As shown, each drive gear **2200d** may be coupled to or integrally formed from a respective worm gear **2200a**. Each drive gear **2200d** is configured to engage a corresponding intermediary gear **2200b** and each intermediary gear is configured to engage a corresponding pad gear **2200c**. The plurality of gears **2200** and the worm **2202** may be generally described as a drivetrain configured to transfer rotational motion from the pad motor **2204** to each of the cleaning pads **2010**.

An example of a steam cleaning apparatus, consistent with the present disclosure, may include a wand, a cleaning assembly coupled to the wand, the cleaning assembly including a steam generator, and a cleaning head pivotally coupled to the wand. The cleaning head may include a plurality of cleaning pads, a pad motor having a motor driveshaft, a drivetrain coupling the pad motor to each of the plurality of cleaning pads such that a rotation of the motor driveshaft causes a corresponding rotation in each of the plurality of cleaning pads, and a temperature sensor positioned proximate to the drivetrain, the temperature sensor being configured to measure a temperature of the drivetrain.

In some instances, the drivetrain may include a worm coupled to the motor driveshaft. In some instances, the

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drivetrain may include a plurality of worm gears configured to engage with the worm, a plurality of intermediary gears each configured to engage with a respective worm gear, and a plurality of pad gears each configured to engage with a respective intermediary gear. In some instances, the temperature sensor may be positioned proximate to the worm. In some instances, each pad gear may be coupled to a corresponding pad holder such that each pad holder rotates with a corresponding pad gear. In some instances, each pad holder may include a steam distribution cavity and a pad driveshaft defining a steam channel that is fluidly coupled to the steam distribution cavity and the cleaning assembly. In some instances, the steam distribution cavity may include a plurality of ribs extending within the steam distribution cavity, the plurality of ribs extending radially outward from a pad rotation axis. In some instances, each rib may include a plurality of rib protrusions that are configured to engage with a corresponding cleaning pad. In some instances, a first plurality of rib protrusions corresponding to a first rib may be radially offset from a plurality of rib protrusions corresponding to an immediately adjacent rib. In some instances, the pad driveshaft may include a coupling configured to couple the pad holder to the drivetrain. In some instances, the coupling may include a coupling body having a first end and a second end, the first end is spaced apart from the second end such that the coupling body partially encloses an area. In some instances, the coupling body may include a plurality of arcuate regions, wherein immediately adjacent arcuate regions are separated by a recessed region. In some instances, each pad holder may include one or more steam outlets fluidly coupled to the steam distribution cavity. In some instances, the one or more steam outlets may be configured to divert steam passing therethrough such that the diverted steam does not pass through the plurality of cleaning pads. In some instances, each cleaning pad may include a pad mount, the pad mount having an annular base and one or more mount protrusions extending from the annular base.

Another example of steam cleaning apparatus, consistent with the present disclosure, may include a wand, a cleaning assembly coupled to the wand, the cleaning assembly including a steam generator, and a cleaning head pivotally coupled to the wand. The cleaning head may include a plurality of cleaning pads, a pad motor having a motor driveshaft, and a drivetrain coupling the pad motor to each of the plurality of cleaning pads such that a rotation of the motor driveshaft causes a corresponding rotation in each of the plurality of cleaning pads, the drivetrain including a drive bevel gear coupled to the motor driveshaft, a plurality of step-down bevel assemblies, and a plurality pad bevel gears.

In some instances, each step-down bevel assembly may include a first step-down bevel gear and a second step-down bevel gear, each coupled to a gear driveshaft, the first step-down bevel gear engaging the drive bevel gear and the second step-down bevel gear engaging the pad bevel gear. In some instances, each gear driveshaft may be supported by one or more thrust bearings. In some instances, each pad bevel gear may be coupled to a corresponding pad holder such that each pad holder rotates with a corresponding pad bevel gear. In some instances, each pad holder may include a steam distribution cavity and a pad driveshaft defining a steam channel that is fluidly coupled to the steam distribution cavity and the cleaning assembly. In some instances, the steam distribution cavity may include a plurality of ribs extending within the steam distribution cavity, the plurality of ribs extending radially outward from a pad rotation axis. In some instances, each rib may include a plurality of rib

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protrusions that are configured to engage with a corresponding cleaning pad. In some instances, a first plurality of rib protrusions corresponding to a first rib may be radially offset from a plurality of rib protrusions corresponding to an immediately adjacent rib. In some instances, the pad drive-
 shaft may include a coupling configured to couple the pad holder to the drivetrain. In some instances, the coupling may include a coupling body having a first end and a second end, the first end is spaced apart from the second end such that the coupling body partially encloses an area. In some instances, the coupling body may include a plurality of arcuate regions, wherein immediately adjacent arcuate regions are separated by a recessed region. In some instances, each pad holder may include one or more steam outlets fluidly coupled to the steam distribution cavity. In some instances, the one or more steam outlets may be configured to divert steam passing therethrough such that the diverted steam does not pass through the plurality of cleaning pads. In some instances, each cleaning pad may include a pad mount, the pad mount having an annular base and one or more mount protrusions extending from the annular base.

An example of a steam cleaning apparatus, consistent with the present disclosure, may include a wand, a cleaning assembly coupled to the wand, the cleaning assembly including a steam generator, and a cleaning head pivotally coupled to the wand. The cleaning head may include a plurality of cleaning pads, a pad motor having a motor driveshaft, a drivetrain configured to transfer rotational motion of the motor driveshaft to each of the plurality of cleaning pads such that a rotation of the motor driveshaft causes a corresponding rotation in each of the plurality of cleaning pads, and a temperature sensor positioned proximate to the drivetrain, the temperature sensor being configured to measure a temperature of the drivetrain.

In some instances, the drivetrain may include a worm coupled to the motor driveshaft. In some instances, the drivetrain may include a plurality of worm gears configured to engage with the worm, a plurality of intermediary gears each configured to engage with a respective worm gear, and a plurality of pad gears each configured to engage with a respective intermediary gear. In some instances, the temperature sensor may be positioned proximate to the worm. In some instances, each pad gear may be coupled to a corresponding pad holder such that each pad holder rotates with a corresponding pad gear. In some instances, each pad holder may include a steam distribution cavity and a pad driveshaft defining a steam channel that is fluidly coupled to the steam distribution cavity and the cleaning assembly. In some instances, the steam distribution cavity may include a plurality of ribs extending within the steam distribution cavity, the plurality of ribs extending radially outward from a pad rotation axis. In some instances, each rib may include a plurality of rib protrusions that are configured to engage with a corresponding cleaning pad. In some instances, a first plurality of rib protrusions corresponding to a first rib may be radially offset from a plurality of rib protrusions corresponding to an immediately adjacent rib. In some instances, the pad driveshaft may include a coupling configured to couple the pad holder to the drivetrain. In some instances, the coupling may include a coupling body having a first end and a second end, the first end is spaced apart from the second end such that the coupling body partially encloses an area. In some instances, the coupling body may include a plurality of arcuate regions, wherein immediately adjacent arcuate regions are separated by a recessed region. In some instances, each pad holder may include one or more steam outlets fluidly coupled to the steam distribution cavity. In

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some instances, the one or more steam outlets may be configured to divert steam passing therethrough such that the diverted steam does not pass through the plurality of cleaning pads. In some instances, each cleaning pad may include a pad mount, the pad mount having an annular base and one or more mount protrusions extending from the annular base. In some instances, the drivetrain may include a drive bevel gear coupled to the motor driveshaft, a plurality of step-down bevel assemblies, and a plurality pad bevel gears and, wherein, each step-down bevel assembly may include a first step-down bevel gear and a second step-down bevel gear, each coupled to a gear driveshaft, the first step-down bevel gear engaging the drive bevel gear and the second step-down bevel gear engaging the pad bevel gear, each gear driveshaft being supported by one or more thrust bearings and each pad bevel gear being coupled to a corresponding pad holder such that each pad holder rotates with a corresponding pad bevel gear.

Another example of a steam cleaning apparatus, consistent with the present disclosure, may include a wand, a cleaning assembly coupled to the wand, the cleaning assembly including a steam generator, and a cleaning head pivotally coupled to the wand. The cleaning head may include a plurality of cleaning pads, a pad motor configured to rotate each of the plurality of cleaning pads, a steam nozzle, and a steam valve fluidly coupled to the cleaning assembly, the steam valve being configured to selectively direct steam passing therethrough to at least one of the steam nozzle or each of the plurality of cleaning pads.

In some instances, the steam valve may be a solenoid valve. In some instances, the steam nozzle may be configured to emit steam along a steam axis, the steam axis intersecting a surface to be cleaned. In some instances, the cleaning head may further include a steam illuminating element configured to illuminate a region of the surface to be cleaned within which the steam axis intersects the surface to be cleaned.

While the principles of the invention have been described herein, it is to be understood by those skilled in the art that this description is made only by way of example and not as a limitation as to the scope of the invention. Other embodiments are contemplated within the scope of the present invention in addition to the exemplary embodiments shown and described herein. Modifications and substitutions by one of ordinary skill in the art are considered to be within the scope of the present invention, which is not to be limited except by the following claims.

What is claimed is:

1. A steam cleaning apparatus comprising:

- a wand;
- a cleaning assembly coupled to the wand, the cleaning assembly including a steam generator; and
- a cleaning head pivotally coupled to the wand, the cleaning head including:
 - a plurality of cleaning pads;
 - a pad motor having a motor driveshaft;
 - a drivetrain configured to transfer rotational motion of the motor driveshaft to each of the plurality of cleaning pads such that a rotation of the motor driveshaft causes a corresponding rotation in each of the plurality of cleaning pads, wherein the drivetrain includes a worm and a plurality of worm gears configured to be rotated by the worm; and
 - a temperature sensor positioned directly above the worm, such that a central axis of the worm is aligned with a central axis of the temperature sensor when the steam cleaning apparatus is in-use to clean a

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floor, the temperature sensor being configured to measure a temperature corresponding to the drivetrain, wherein, when the temperature exceeds a temperature threshold, the pad motor is disabled, the temperature threshold being based, at least in part, on a temperature sensitivity of a material of at least one of the worm gears.

2. The steam cleaning apparatus of claim 1, wherein the worm is coupled to the motor driveshaft.

3. The steam cleaning apparatus of claim 2, wherein the drivetrain includes a plurality of intermediary gears each configured to engage with a respective worm gear and a plurality of pad gears each configured to engage with a respective intermediary gear.

4. The steam cleaning apparatus of claim 3, wherein each pad gear is coupled to a corresponding pad holder such that each pad holder rotates with a corresponding pad gear.

5. The steam cleaning apparatus of claim 4, wherein each pad holder includes a steam distribution cavity and a pad driveshaft defining a steam channel that is fluidly coupled to the steam distribution cavity and the cleaning assembly.

6. The steam cleaning apparatus of claim 5, wherein the steam distribution cavity includes a plurality of ribs extending within the steam distribution cavity, the plurality of ribs extending radially outward from a pad rotation axis.

7. The steam cleaning apparatus of claim 6, wherein each rib includes a plurality of rib protrusions that are configured to engage with a corresponding cleaning pad.

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8. The steam cleaning apparatus of claim 7, wherein a first plurality of rib protrusions corresponding to a first rib are radially offset from a second plurality of rib protrusions corresponding to an immediately adjacent rib.

9. The steam cleaning apparatus of claim 5, wherein the pad driveshaft includes a coupling configured to couple the pad holder to the drivetrain.

10. The steam cleaning apparatus of claim 9, wherein the coupling includes a coupling body having a first end and a second end, the first end is spaced apart from the second end such that the coupling body partially encloses an area.

11. The steam cleaning apparatus of claim 10, wherein the coupling body includes a plurality of arcuate regions, wherein immediately adjacent arcuate regions are separated by a recessed region.

12. The steam cleaning apparatus of claim 5, wherein each pad holder includes one or more steam outlets fluidly coupled to the steam distribution cavity.

13. The steam cleaning apparatus of claim 12, wherein the one or more steam outlets are configured to divert steam passing therethrough such that the diverted steam does not pass through the plurality of cleaning pads.

14. The steam cleaning apparatus of claim 1, wherein each cleaning pad includes a pad mount, the pad mount having an annular base and one or more mount protrusions extending from the annular base.

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